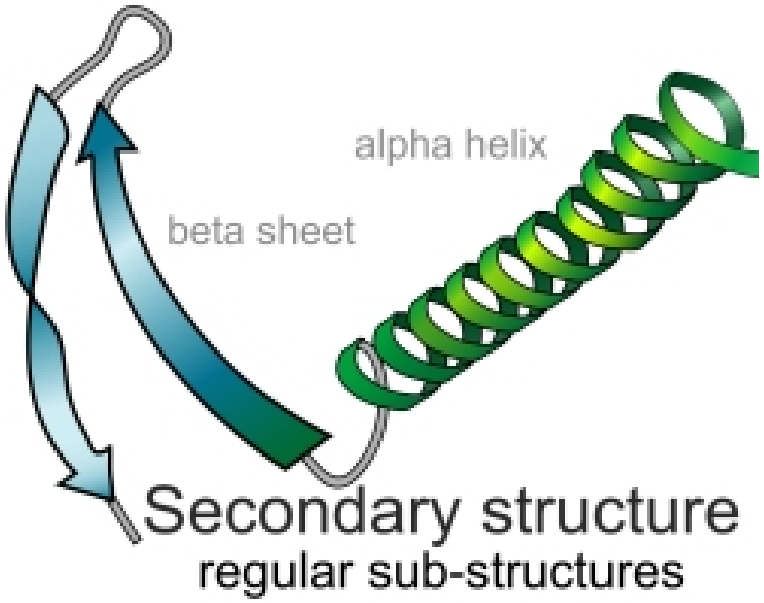
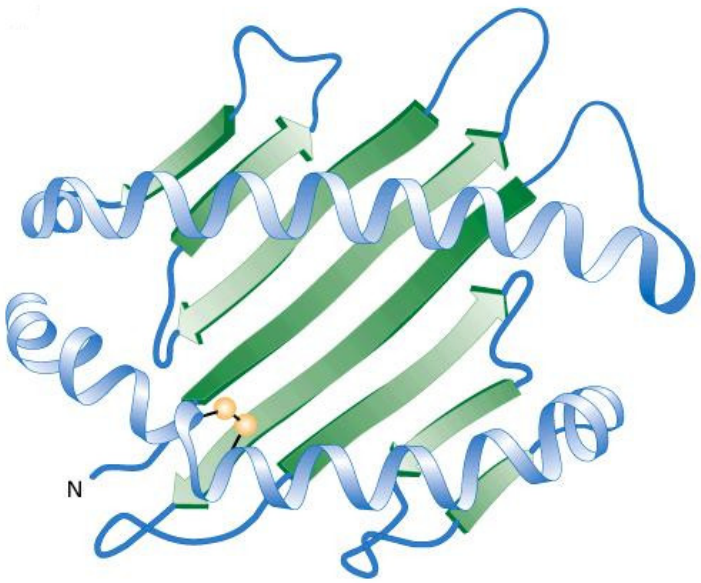
Lecture 9: Neural Networks for Biological Prediction

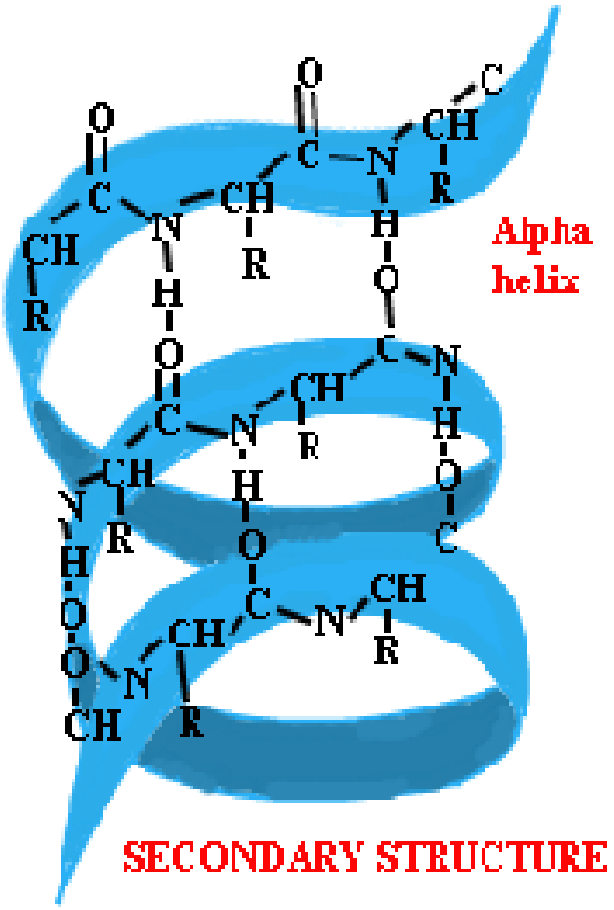




MLP, One-Hot Vectors, Softmax, Profile, Psi-Blast

Outline

1. **MLP**
2. Secondary Protein Structure
3. Designing MLPs



Neural Networks in Biology

- Neural networks (and support vector machines) are frequently used to make prediction in biology
- We are not going into detail about neural networks (many of you have done this in machine learning)
- However, how to use them in biology is quite subtle and is the main topic of this lecture
- We take as a case study the classic problem of the prediction of secondary structure of proteins
- Although, some of this is dated, the lessons are timeless

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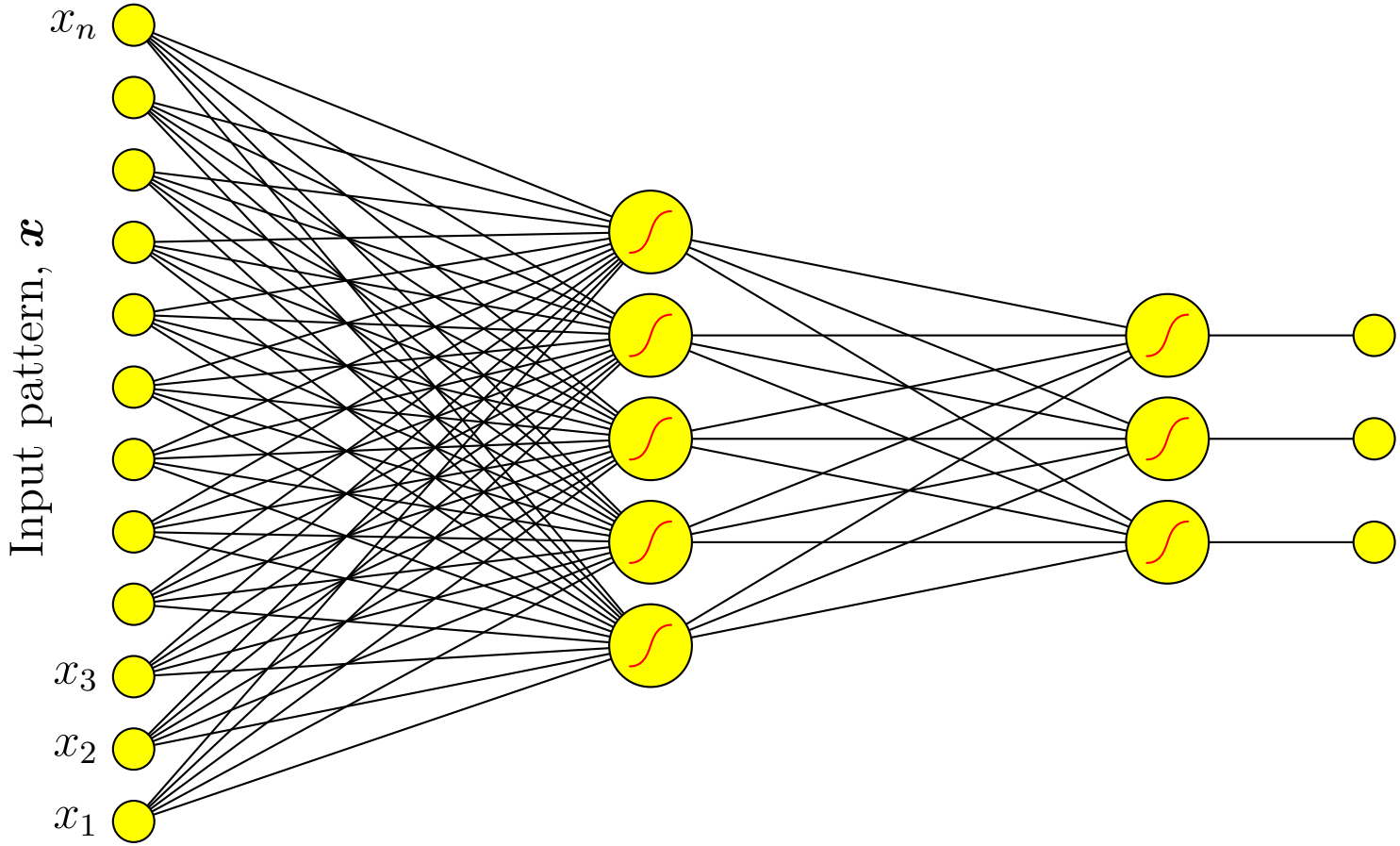

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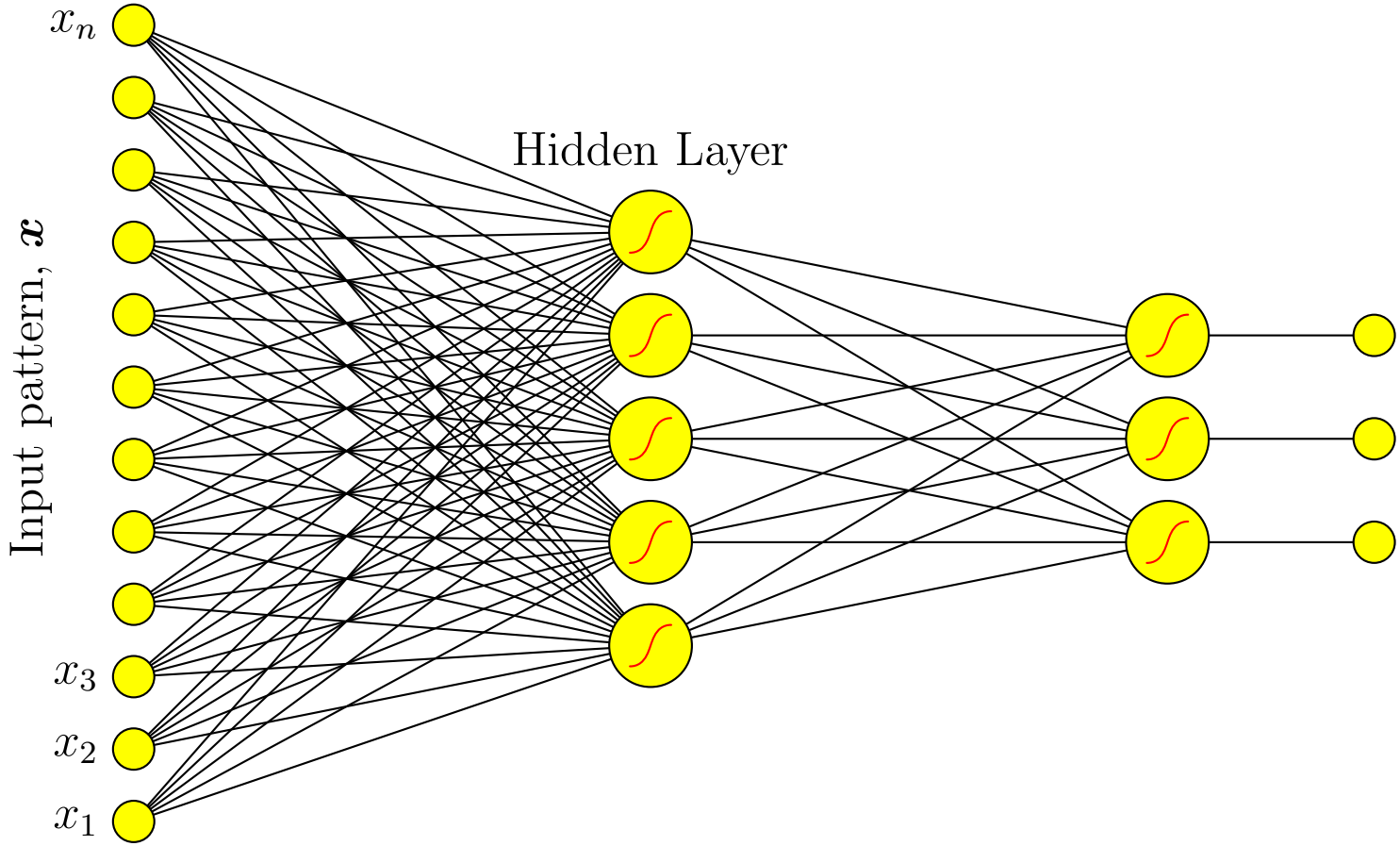

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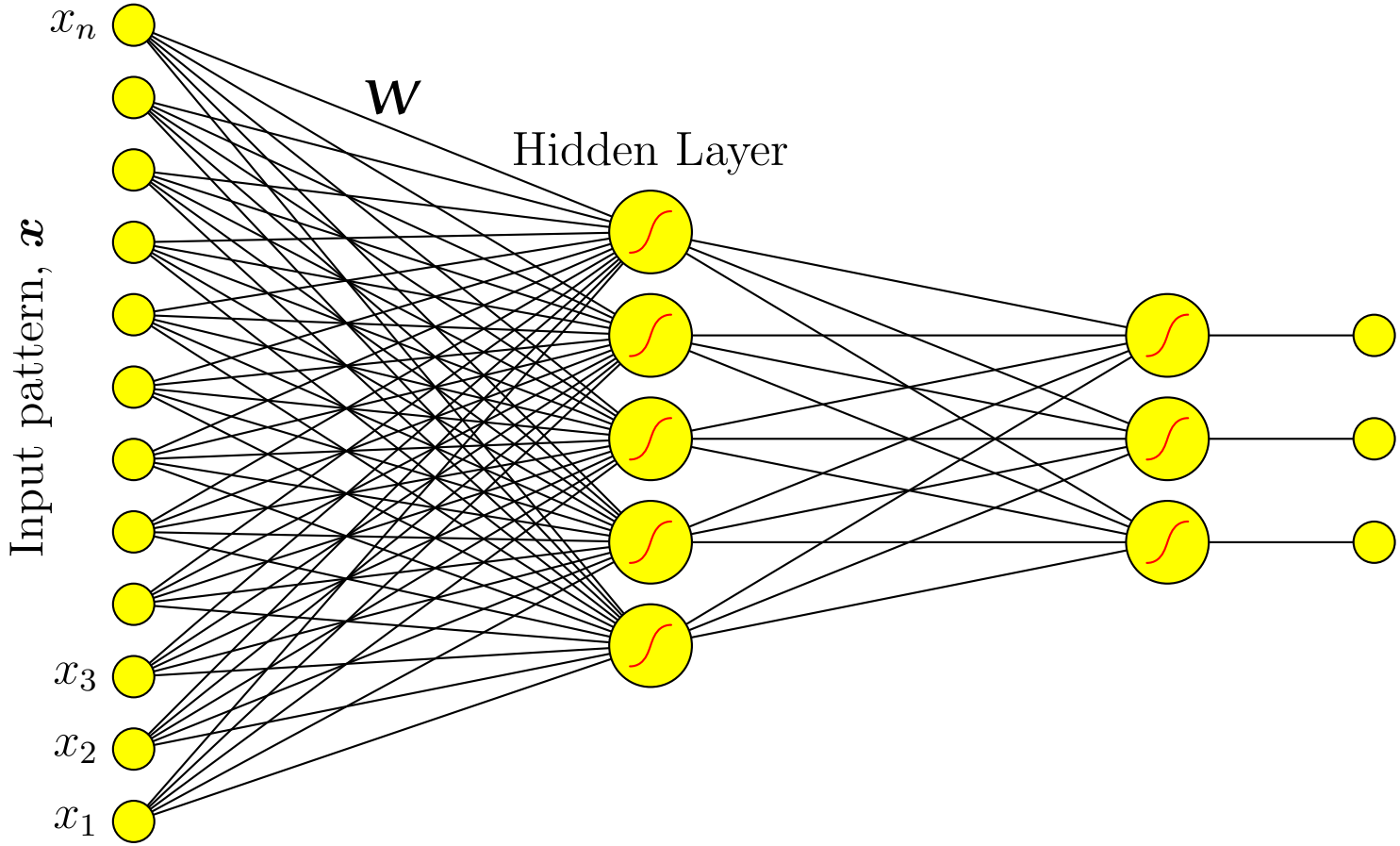

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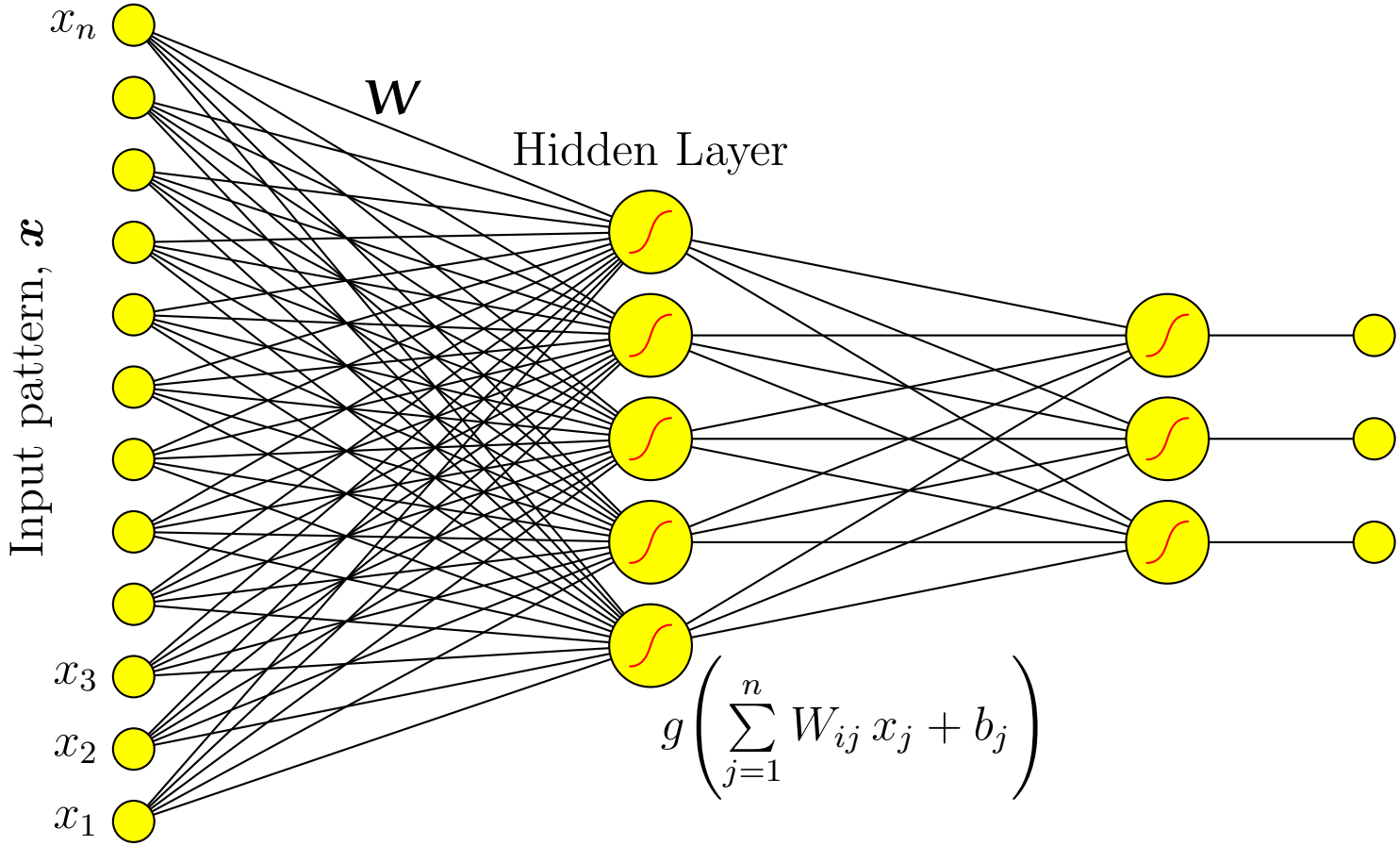

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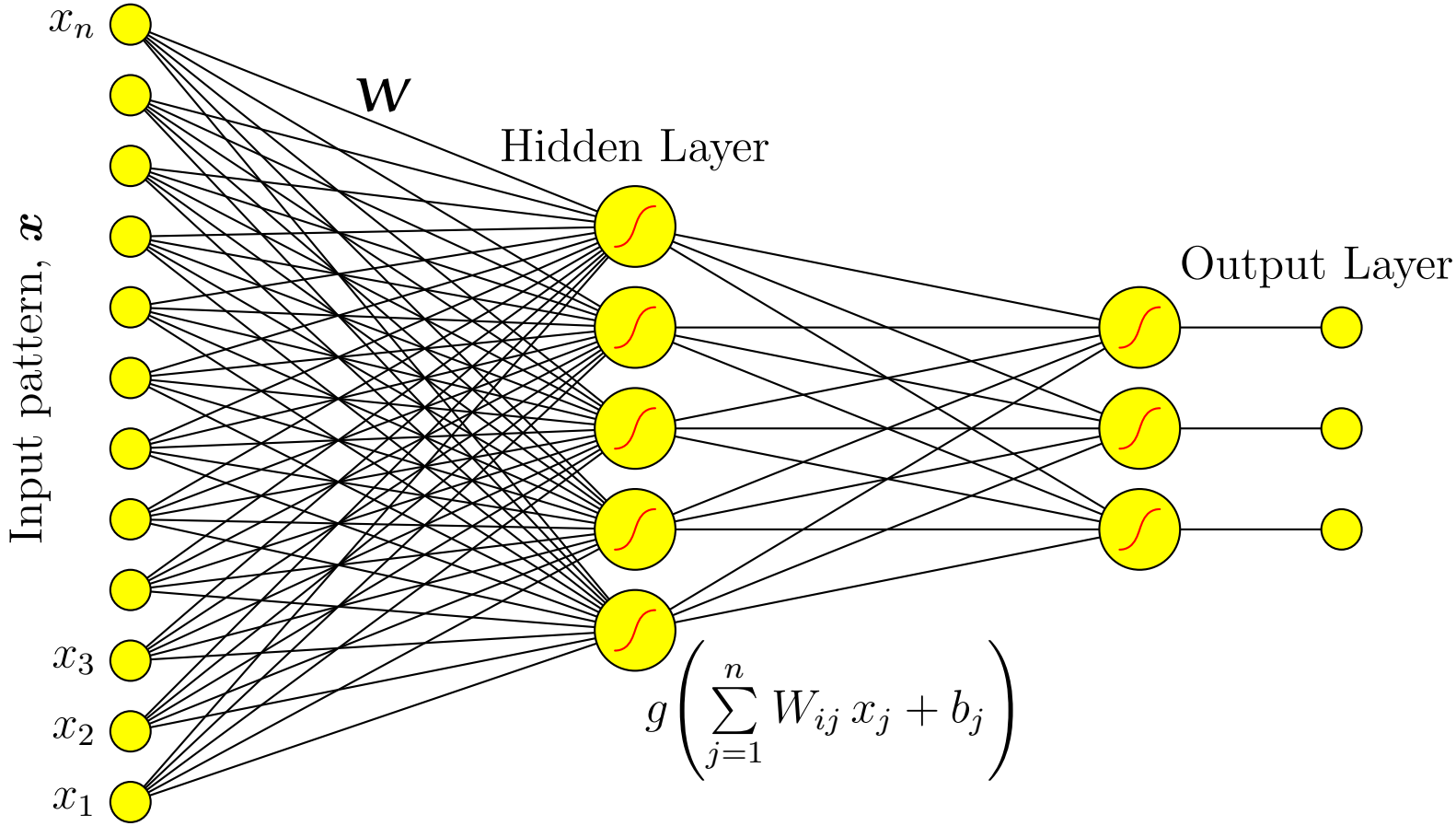

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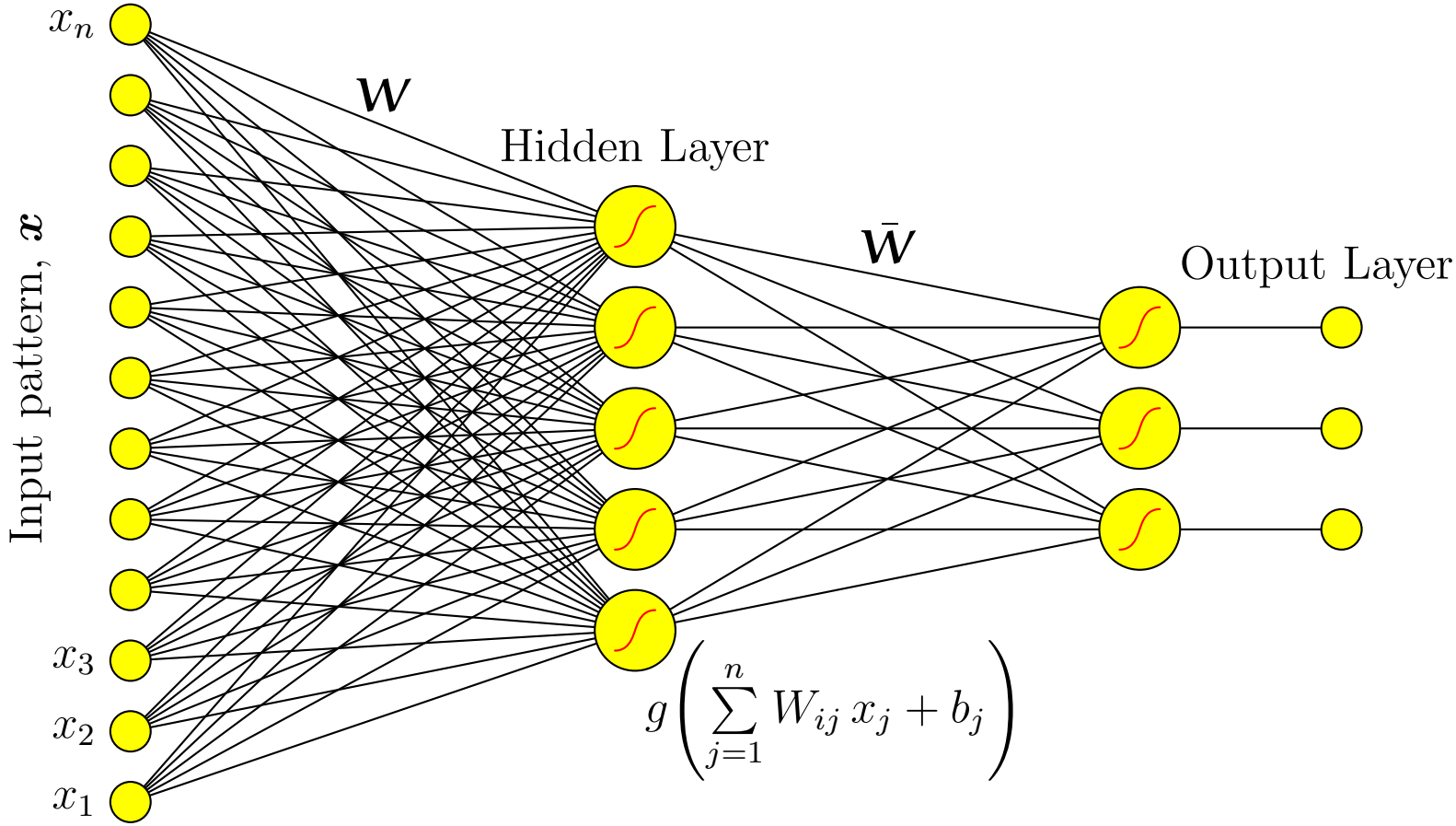

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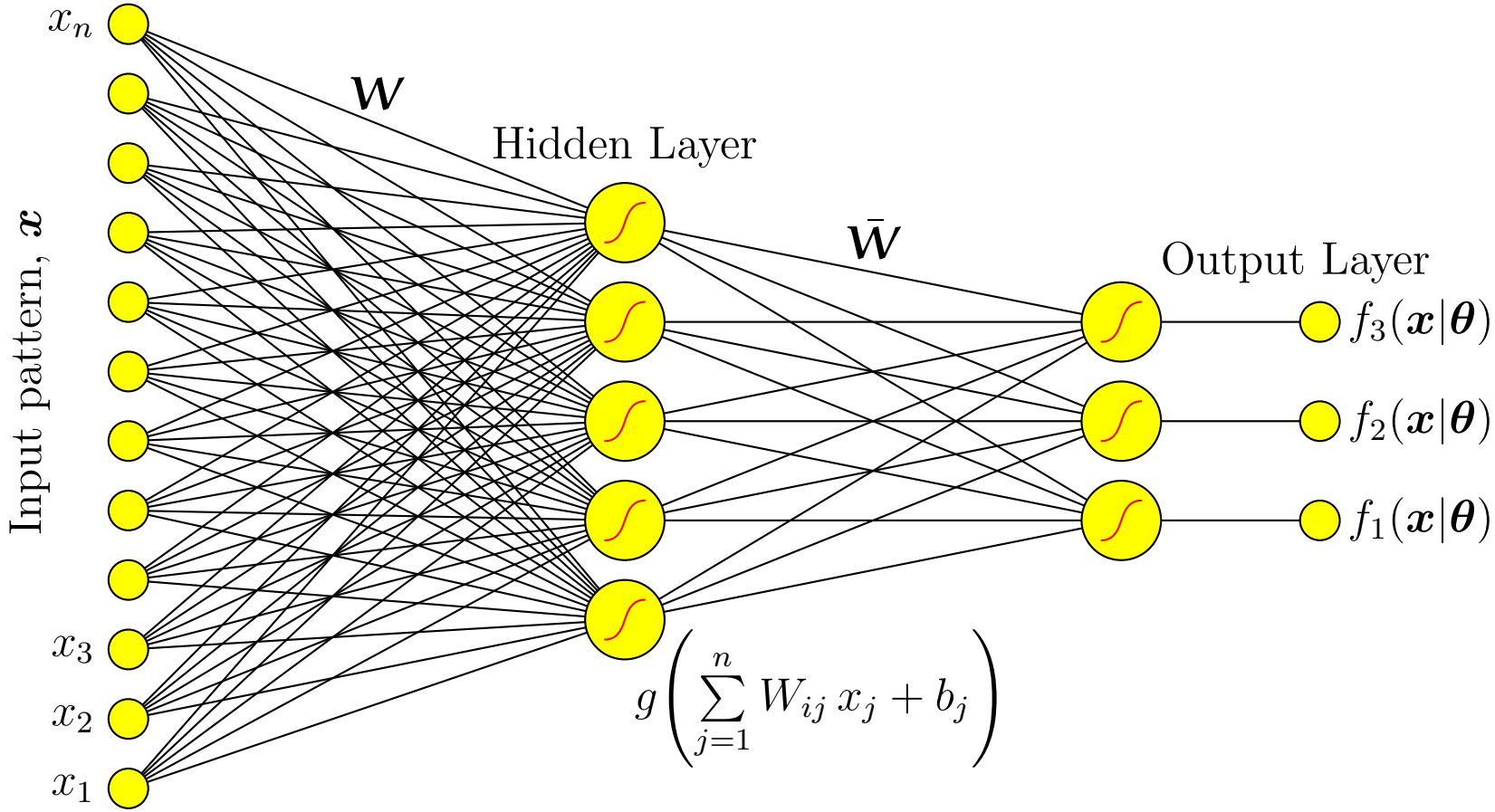

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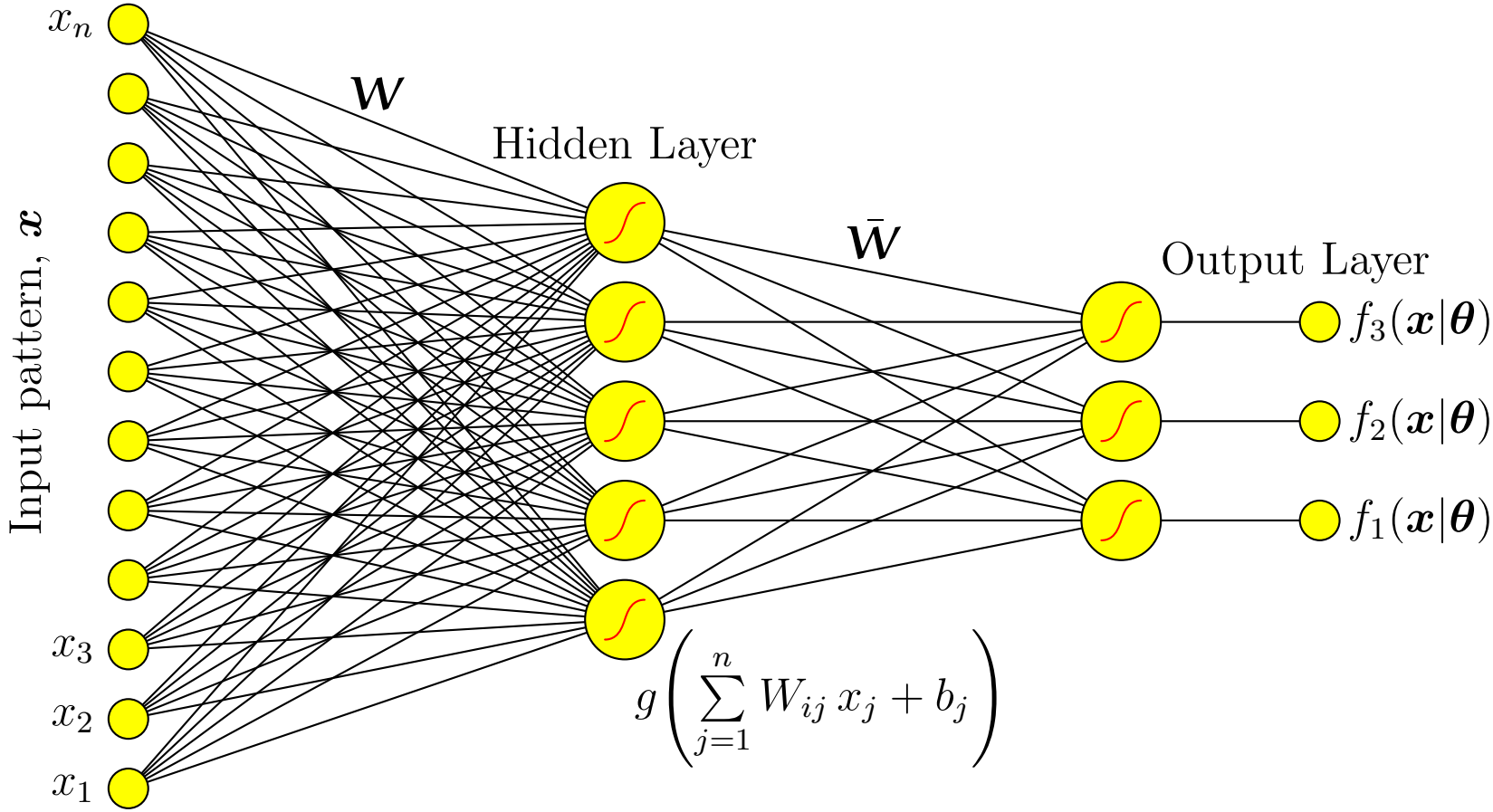

$$f_i(\boldsymbol{x}|\boldsymbol{\theta}) = g\left(\sum_{j=1}^{n_h} \bar{W}_{i,j}\, g\left(\sum_{k=1}^{n} W_{ik}\, x_k + b_k\right) + \bar{b}_i\right)$$

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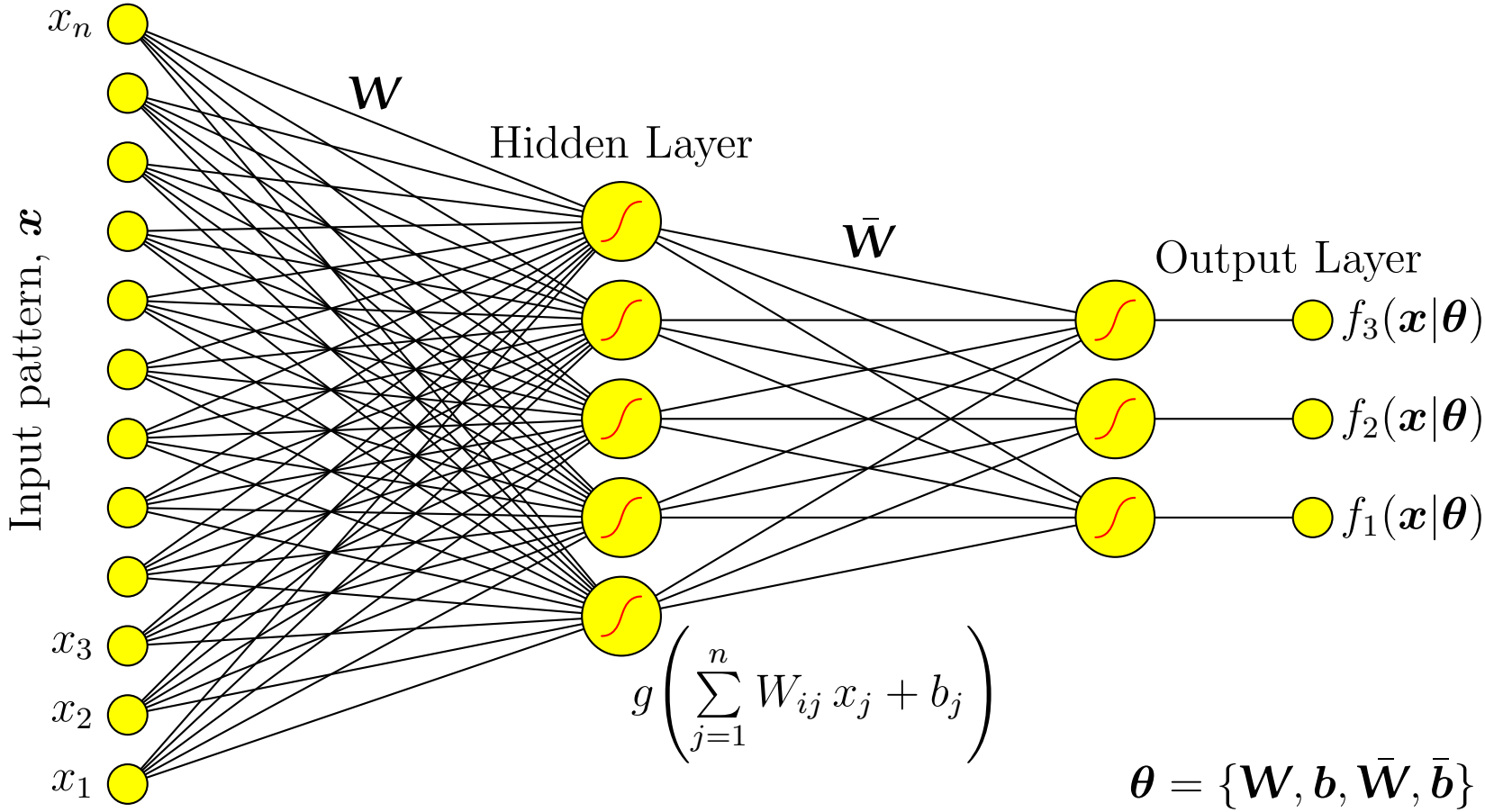

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Training Neural Networks

- We train a neural network on a set of data

$$\mathcal{D} = \left((\boldsymbol{x}^\mu, \boldsymbol{y}^\mu) \,\middle|\, \mu = 1,\, 2,\, \ldots,\, m \right)$$

- Define an loss function $L(\boldsymbol{\theta}, \mathcal{D})$

- E.g. squared error

$$L(\boldsymbol{\theta}, \mathcal{D}) = \frac{1}{m} \sum_{\mu=1}^{m} \left\| \boldsymbol{y}^\mu - \boldsymbol{f}(\boldsymbol{x}^\mu | \boldsymbol{\theta}) \right\|^2$$

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Gradient Descent

- To minimise the loss we follow the negative gradient

$$\boldsymbol{\nabla} L(\boldsymbol{\theta}, \mathcal{D}) = \begin{pmatrix} \frac{\partial L(\boldsymbol{\theta}, \mathcal{D})}{\partial \theta_1} \\ \frac{\partial L(\boldsymbol{\theta}, \mathcal{D})}{\partial \theta_2} \\ \vdots \\ \frac{\partial L(\boldsymbol{\theta}, \mathcal{D})}{\partial \theta_N} \end{pmatrix}$$

- where $\theta_1 = W_{11}$, $\theta_2 = W_{12}$, etc.

- We make a step in the direction of the negative gradient

$$\boldsymbol{\theta}(t+1) = \boldsymbol{\theta}(t) - r\, \boldsymbol{\nabla} L(\boldsymbol{\theta}, \mathcal{D})$$

- r is learning rate

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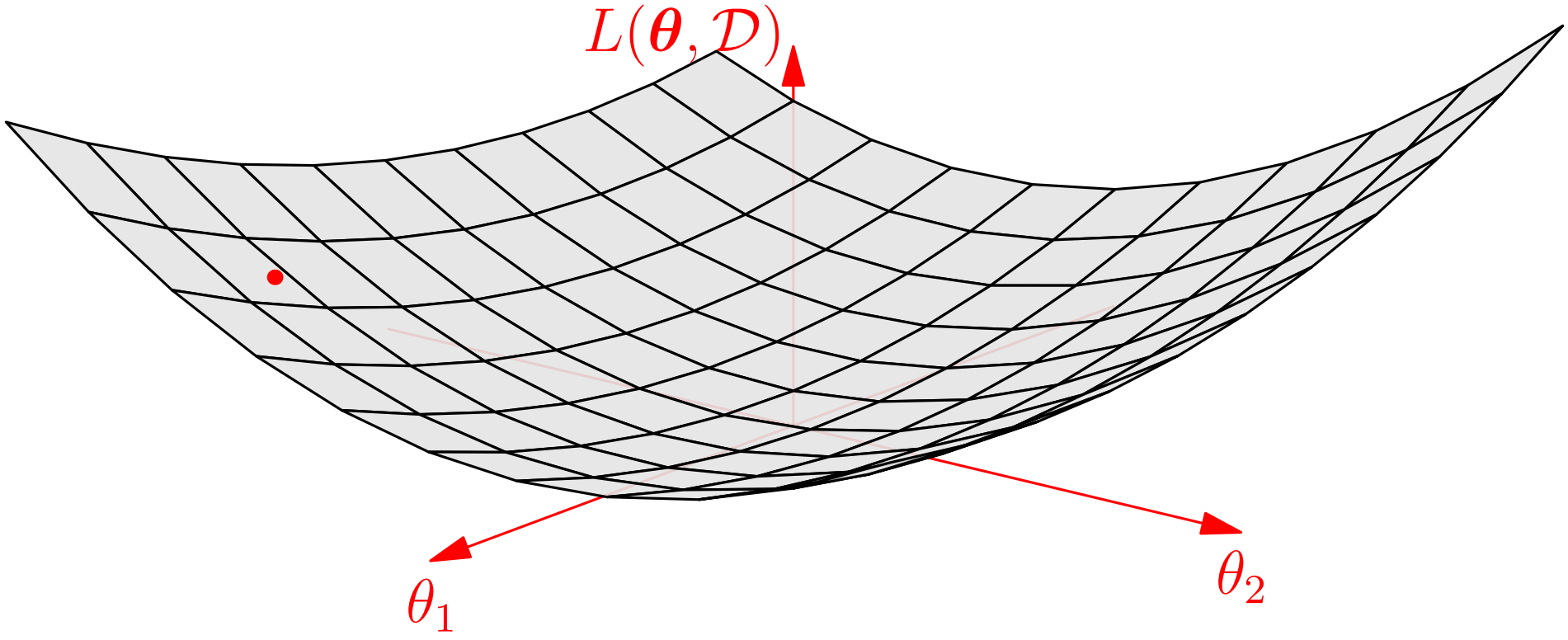

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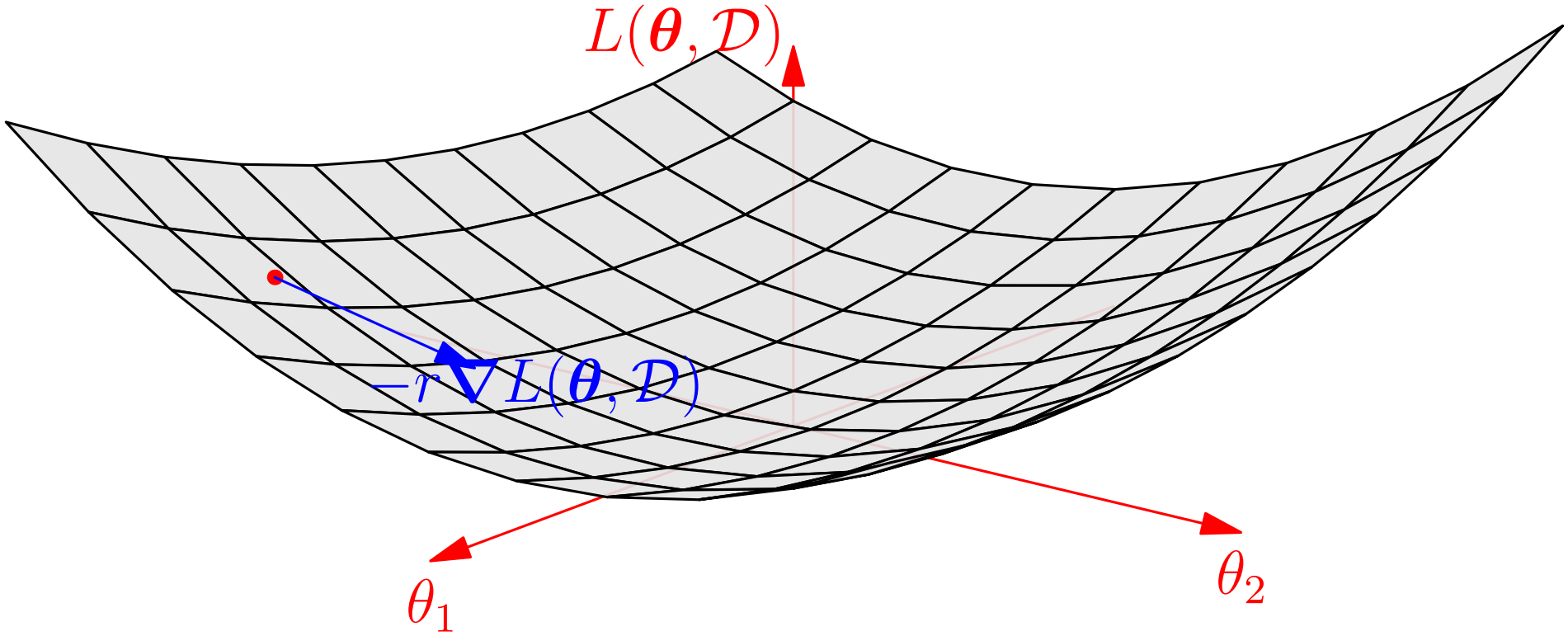

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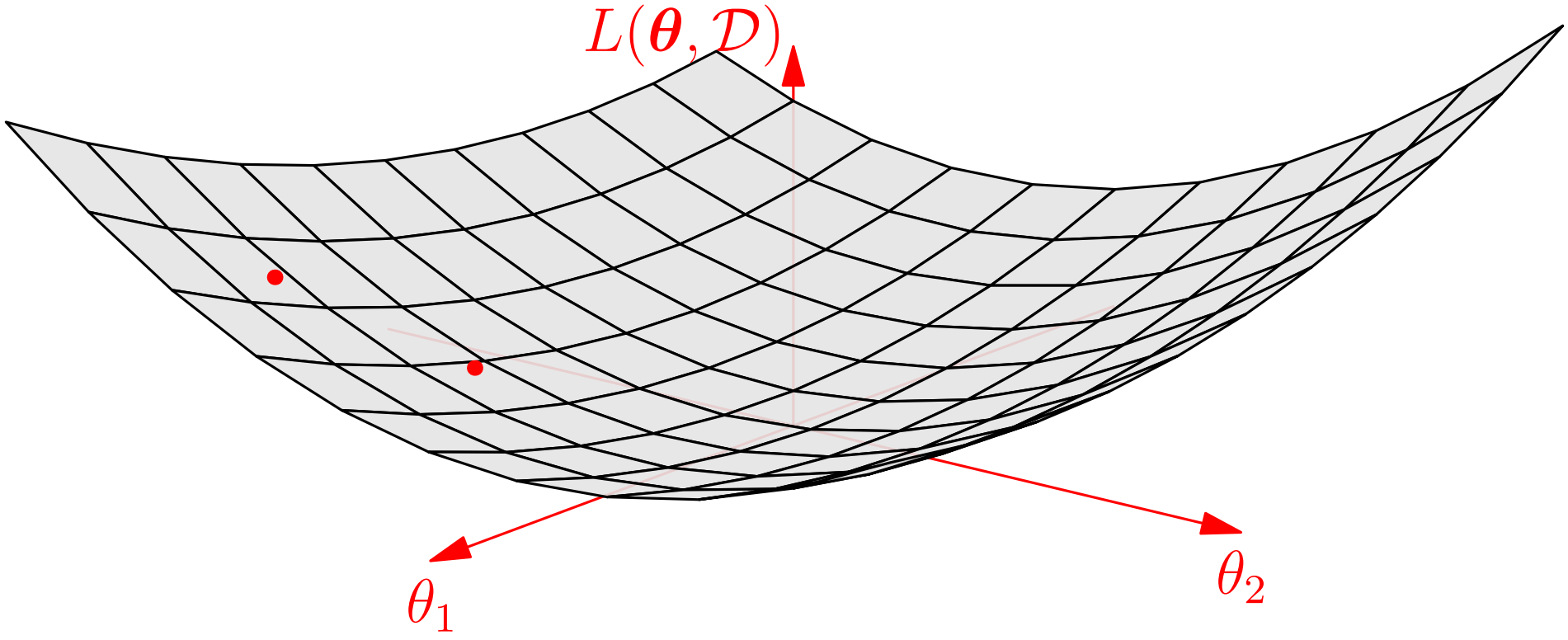

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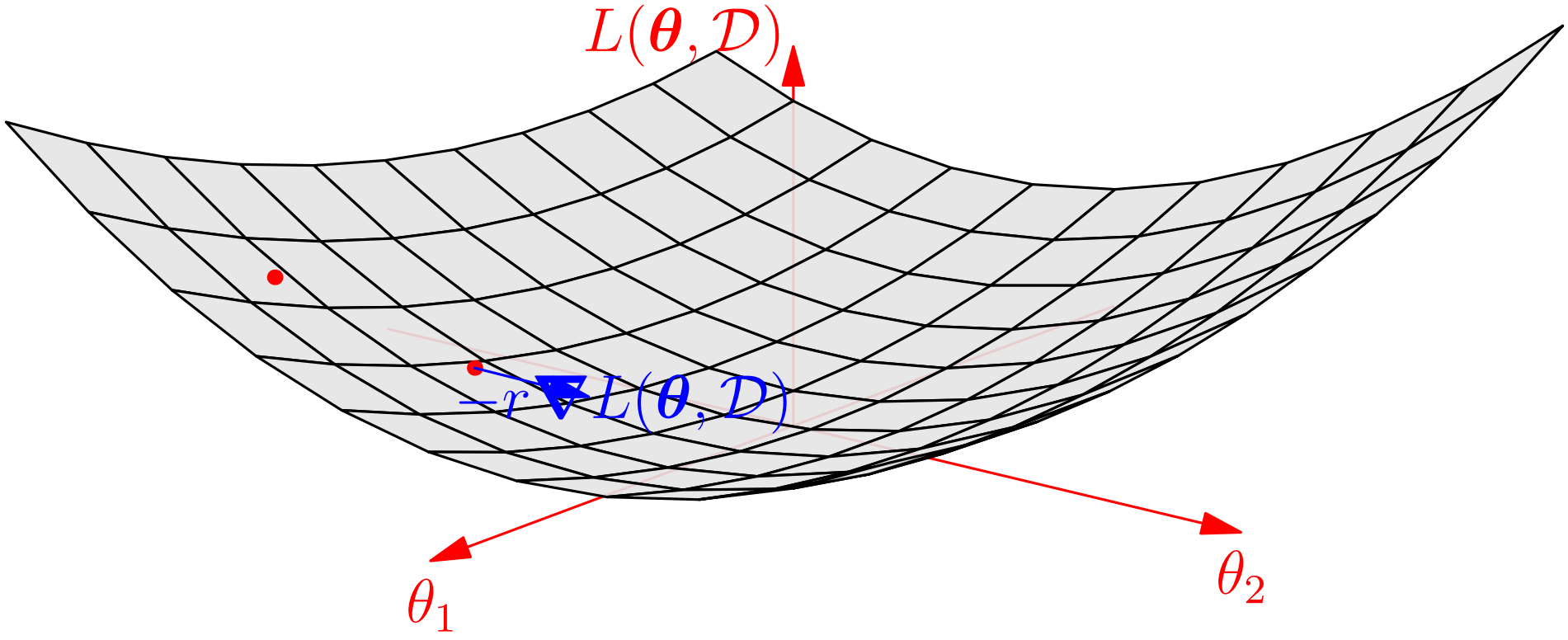

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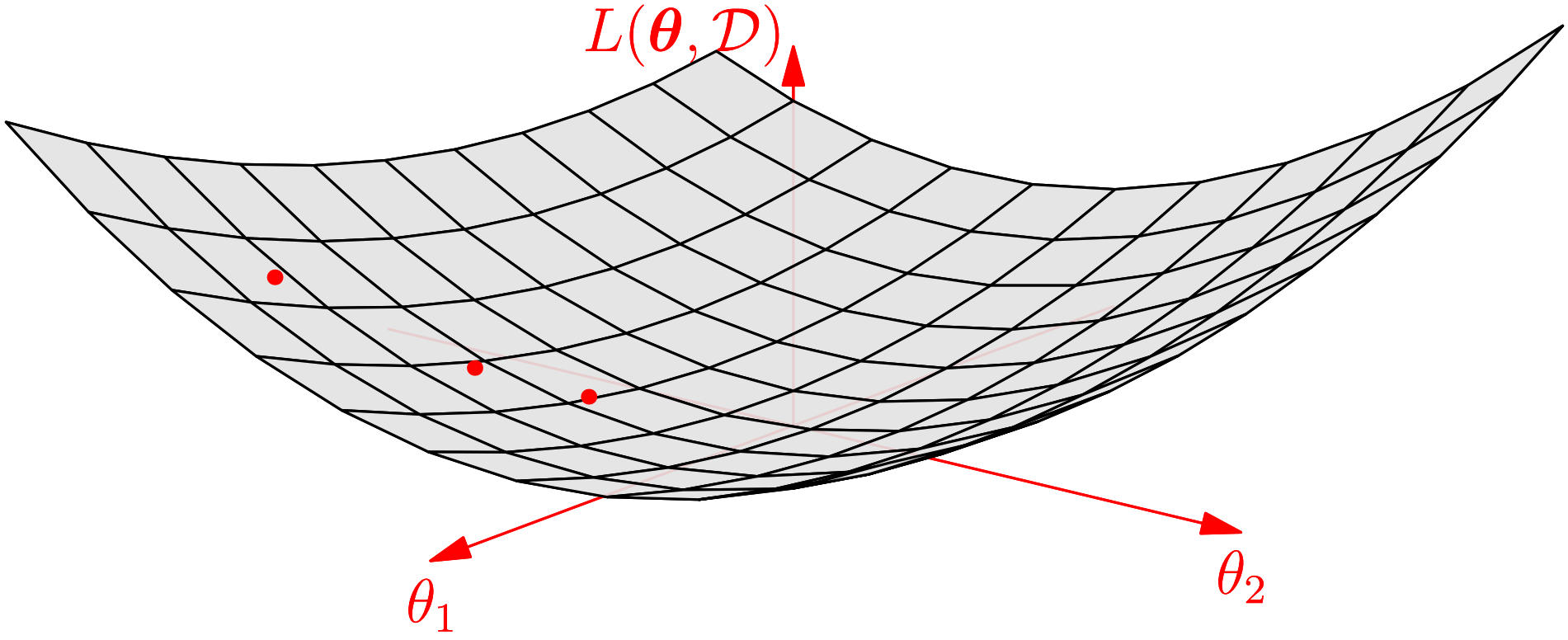

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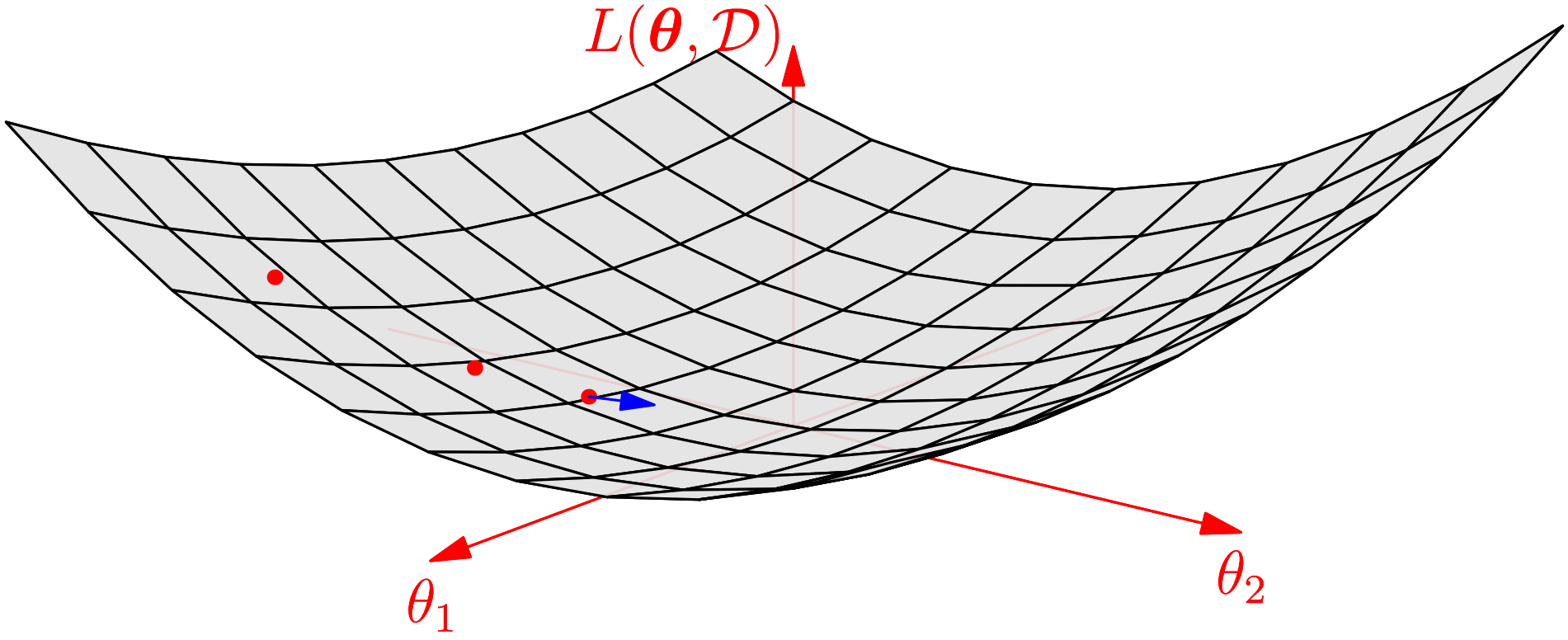

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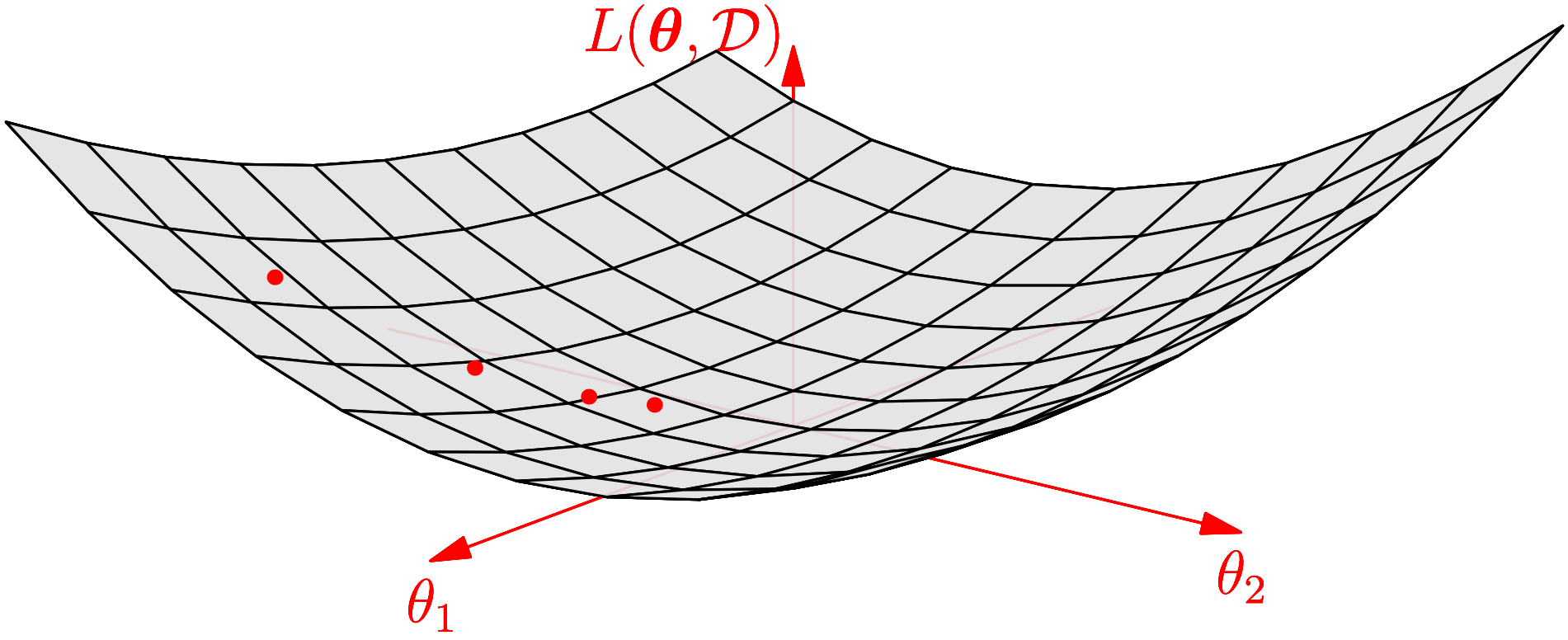

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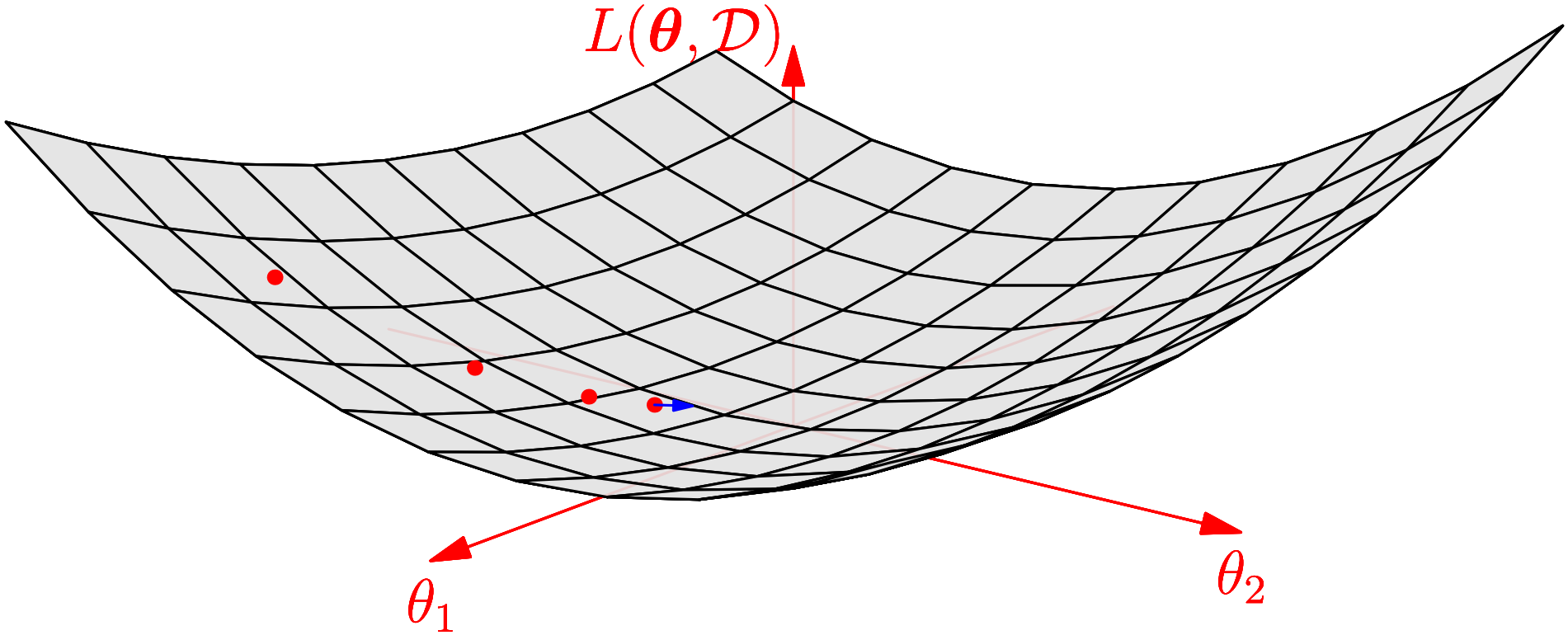

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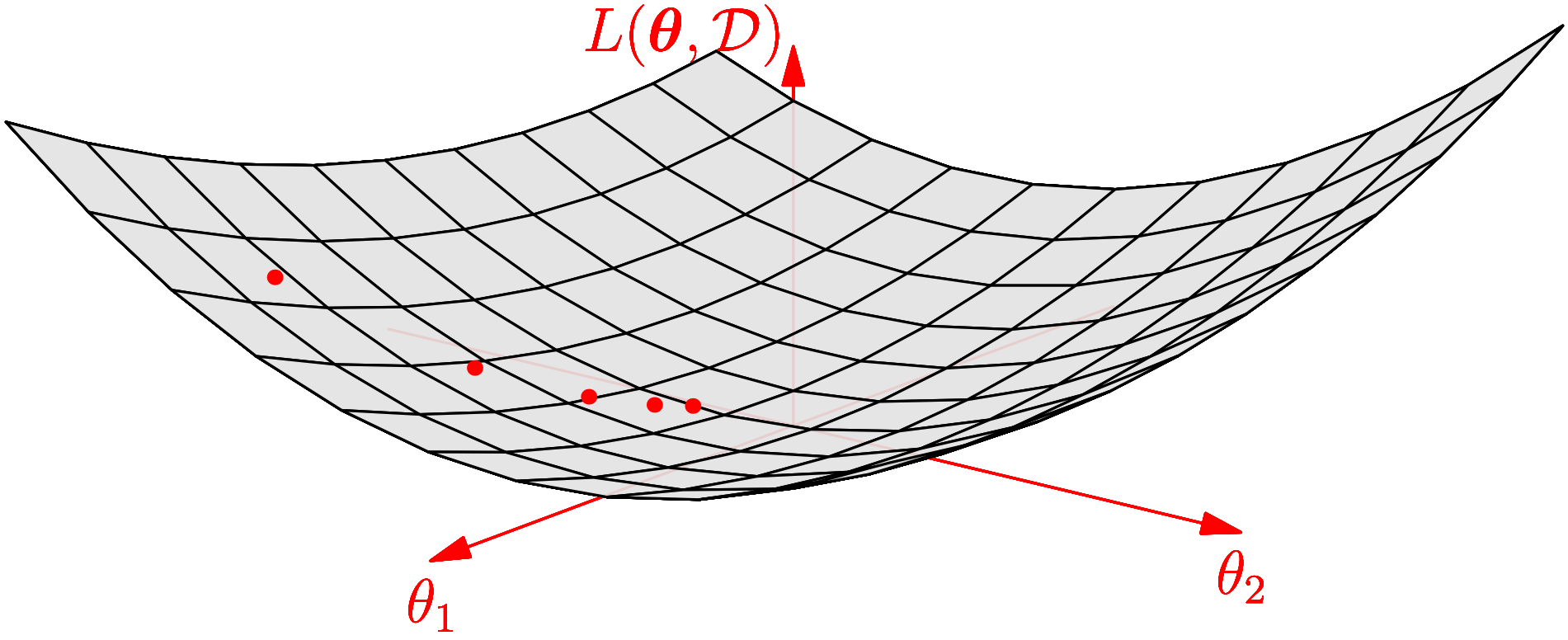

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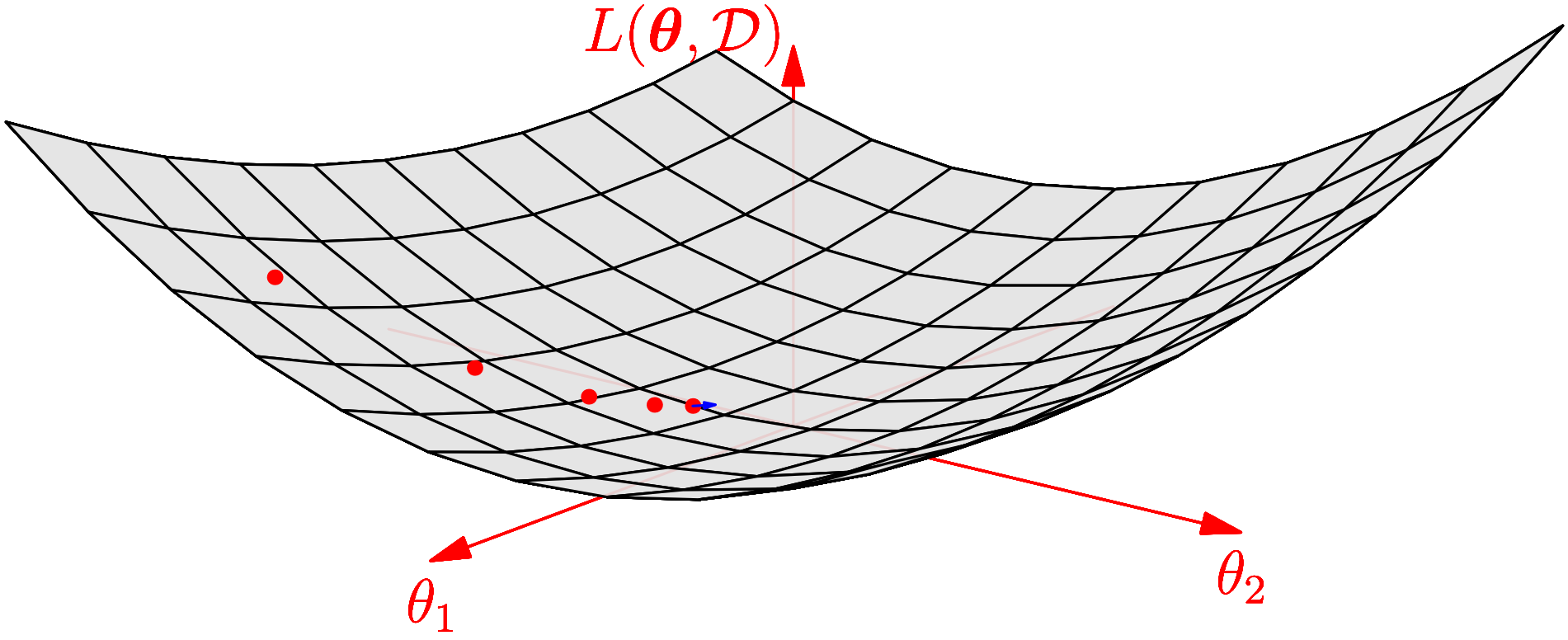

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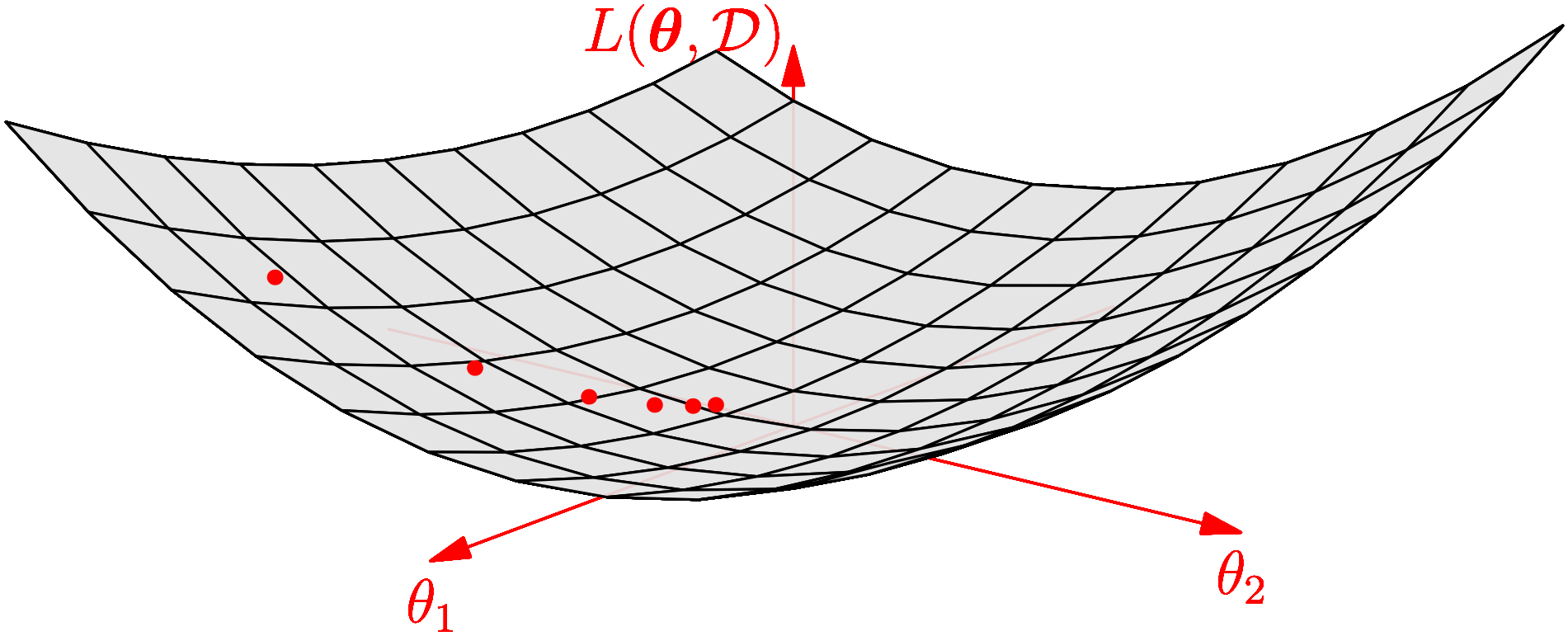

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- Calculating the gradient for a MLP is known as back-propagation
- We can compute the gradient of the error for the whole training set
- In this case we often use Levenberg-Marquardt, scale conjugate gradient or BFGS to speed up convergence
- More often it is effective to find the error per input pattern or mini-batch (a small number of input patterns)
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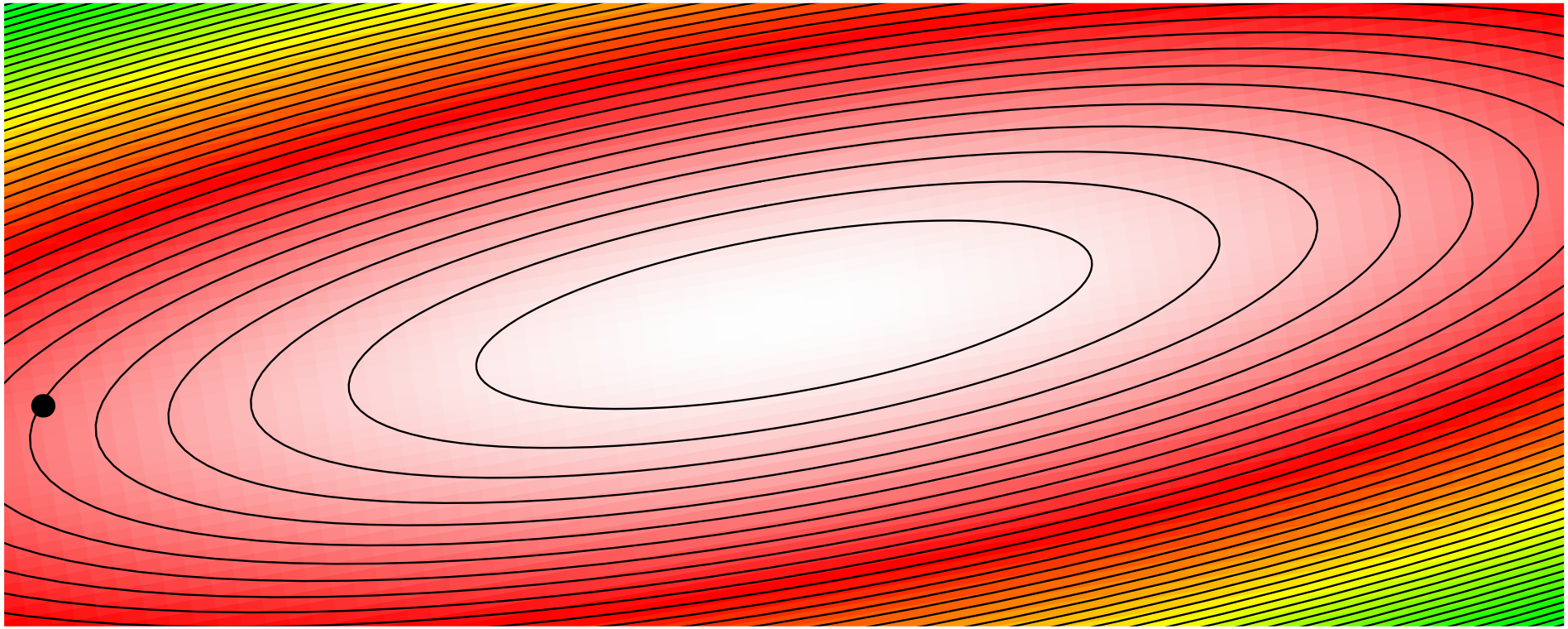
- In higher dimensions the problem is that there are some directions you need to move a long way



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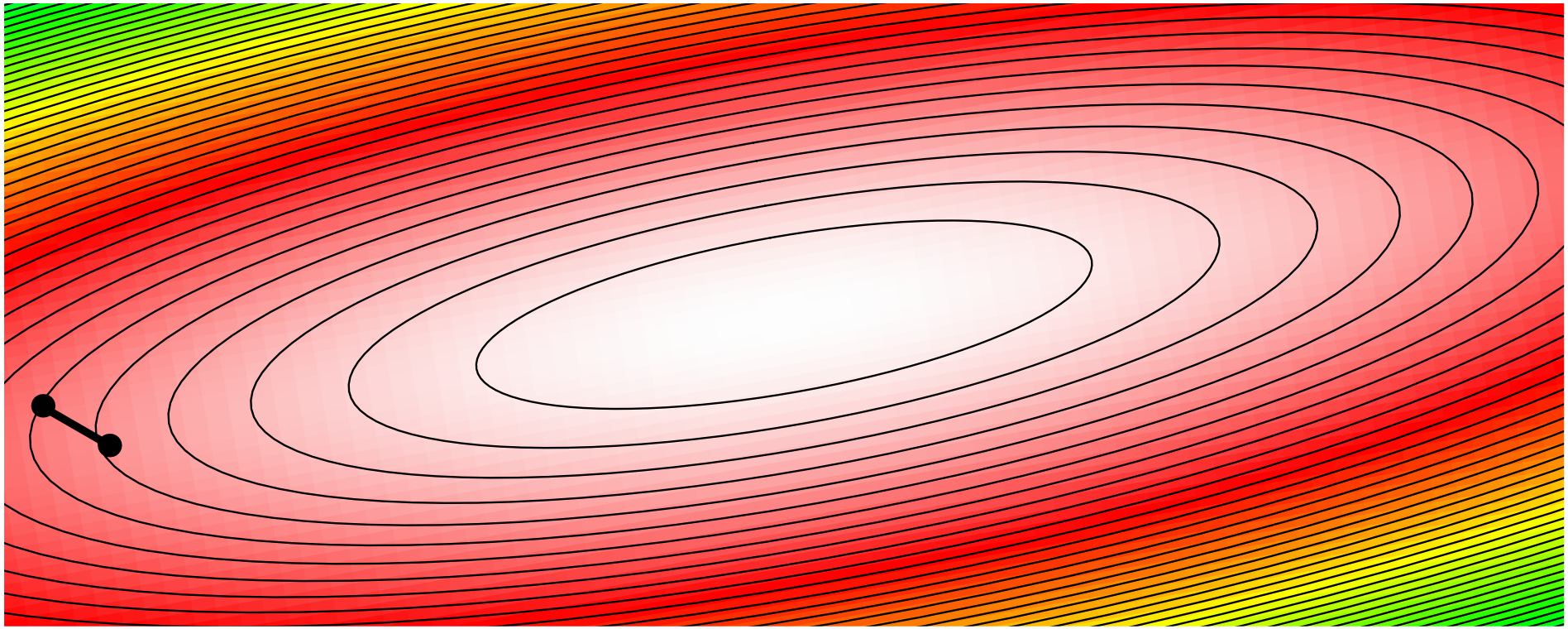

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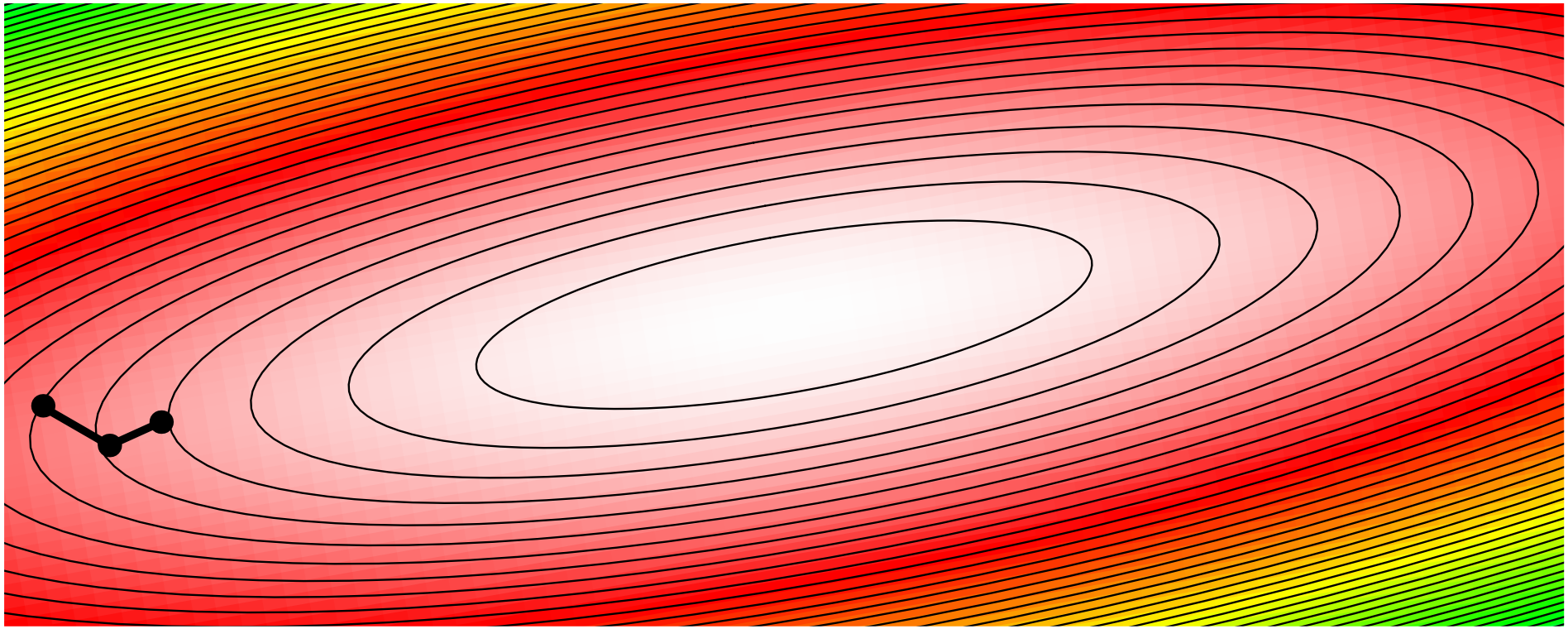

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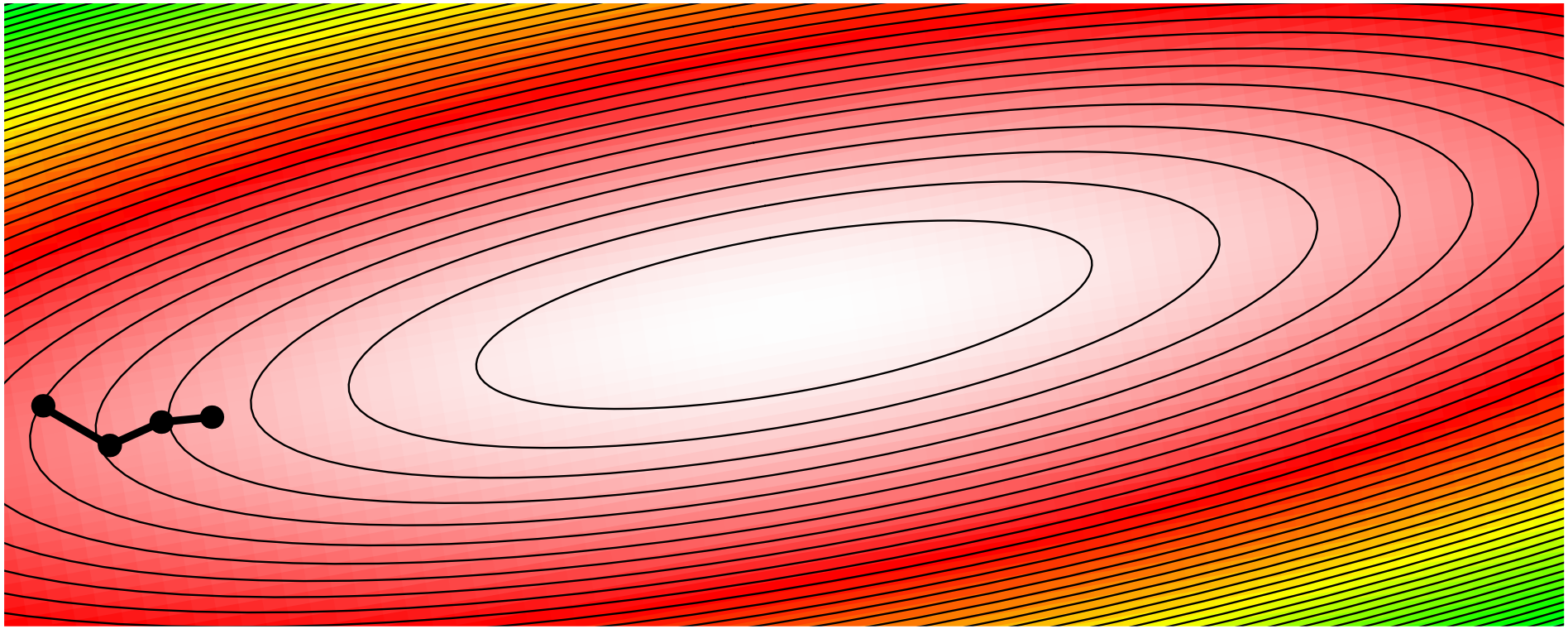

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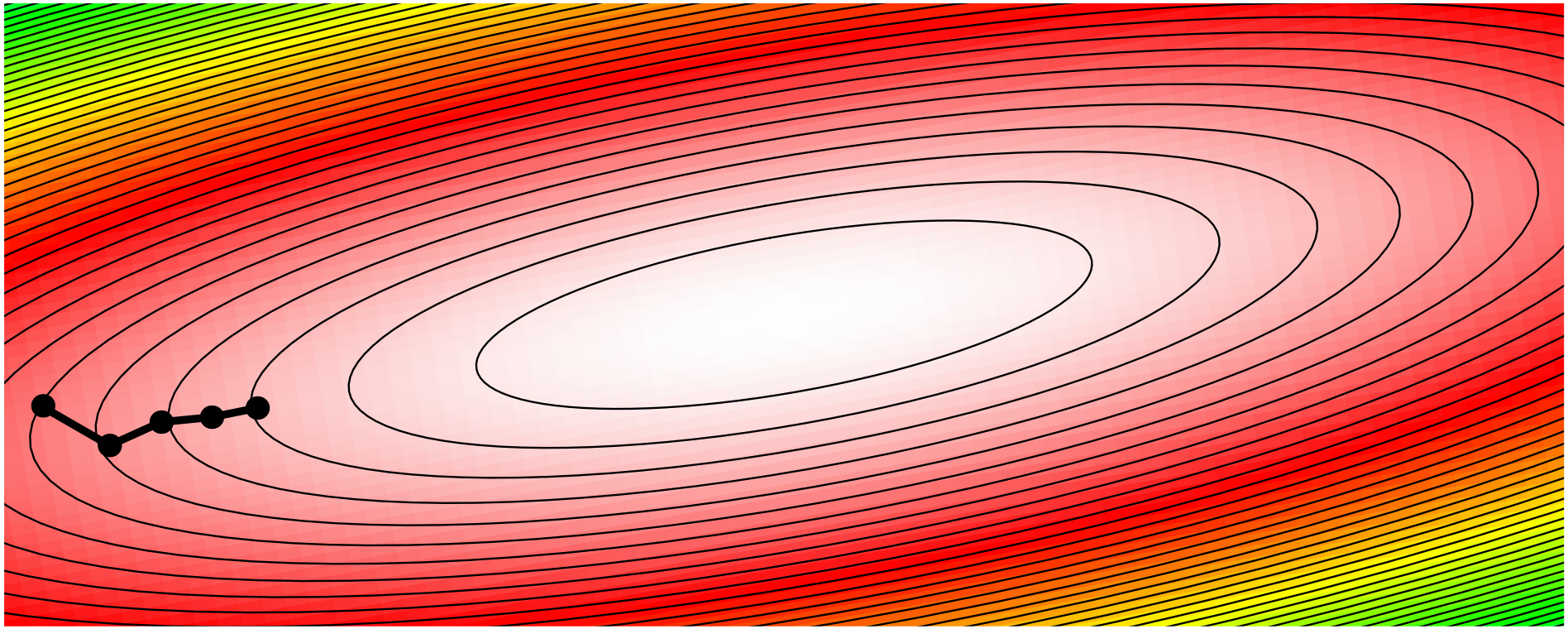

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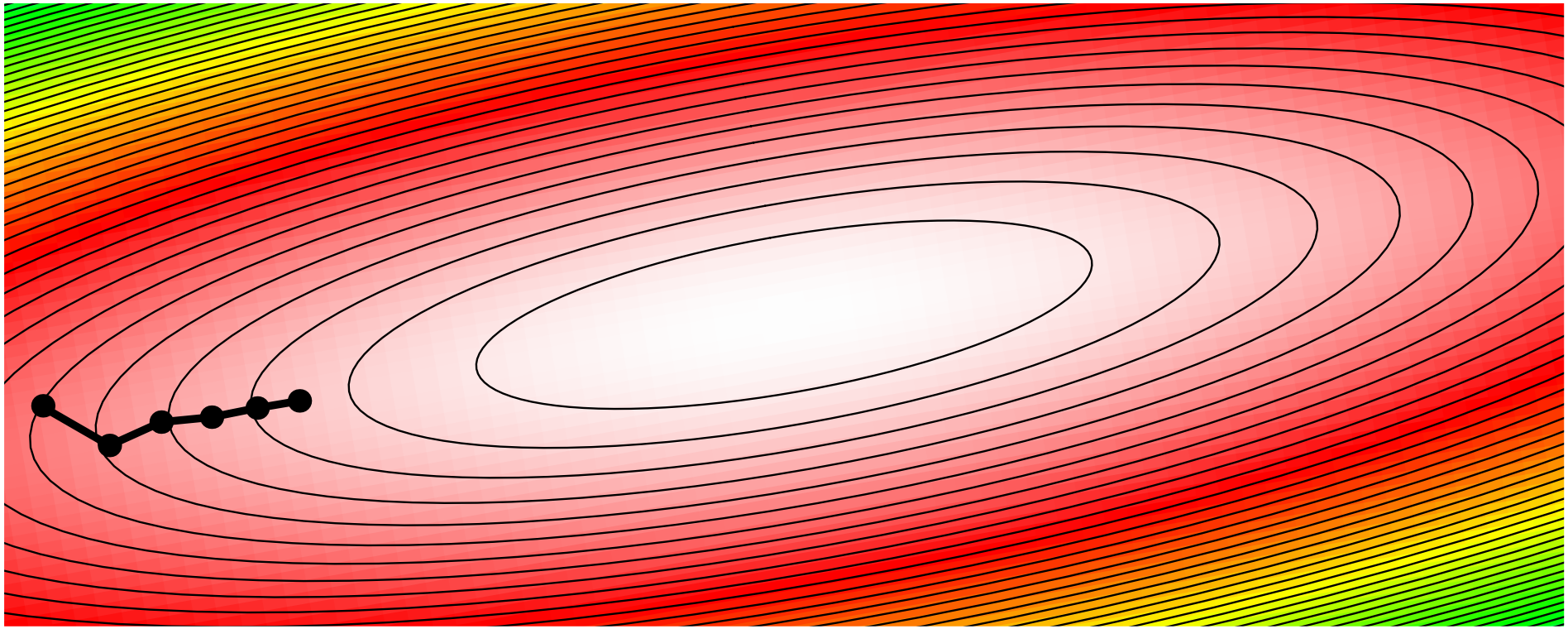

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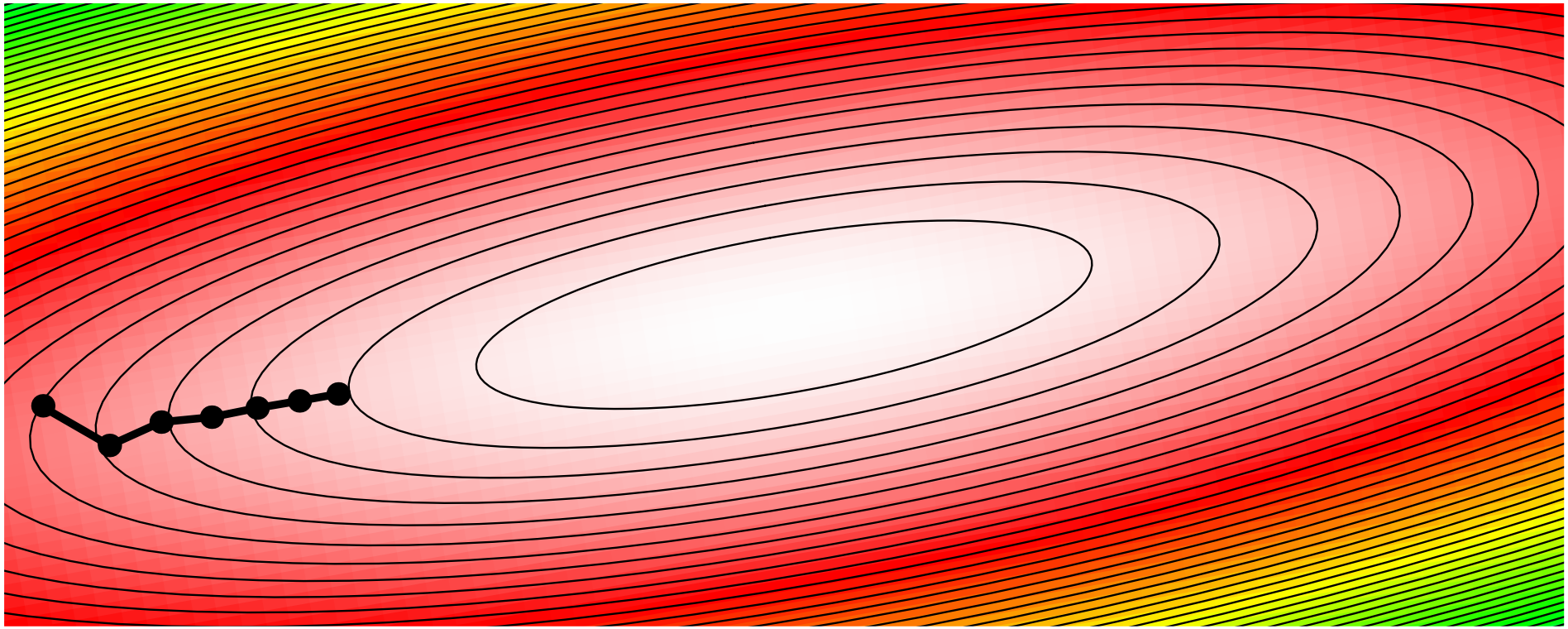

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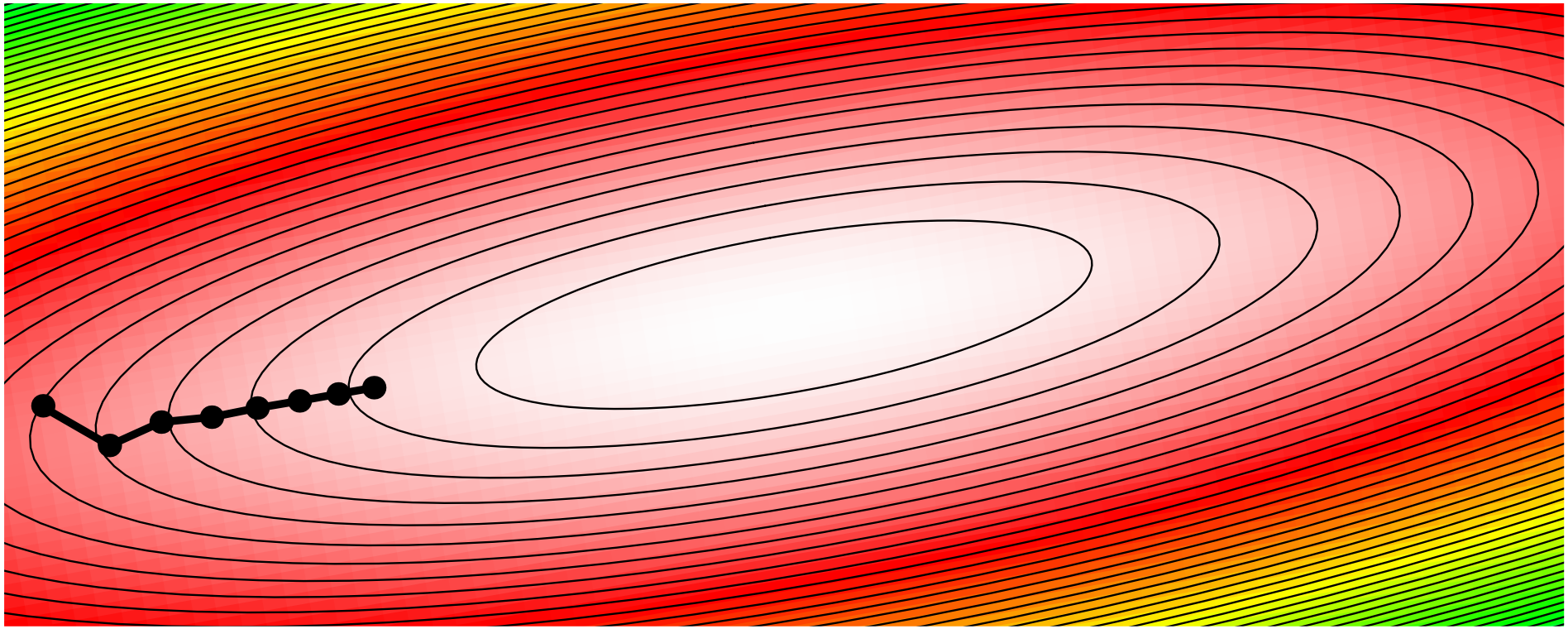

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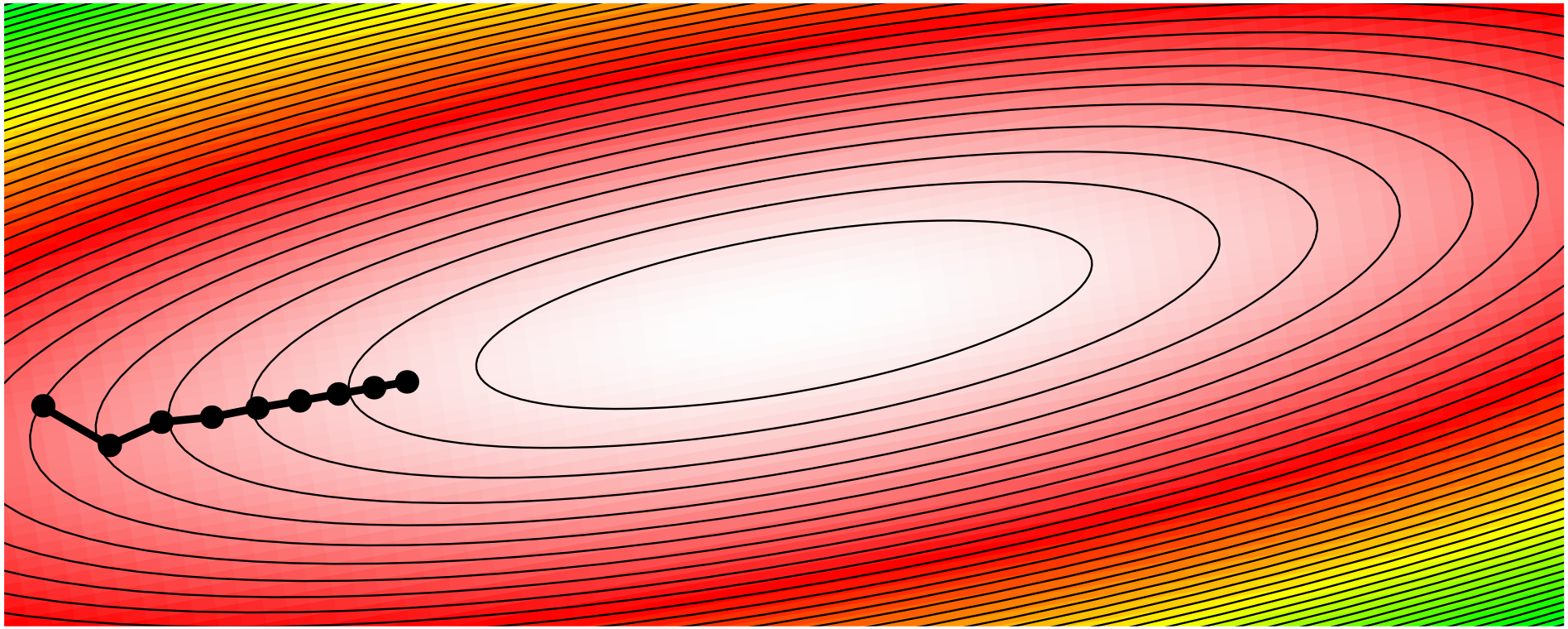

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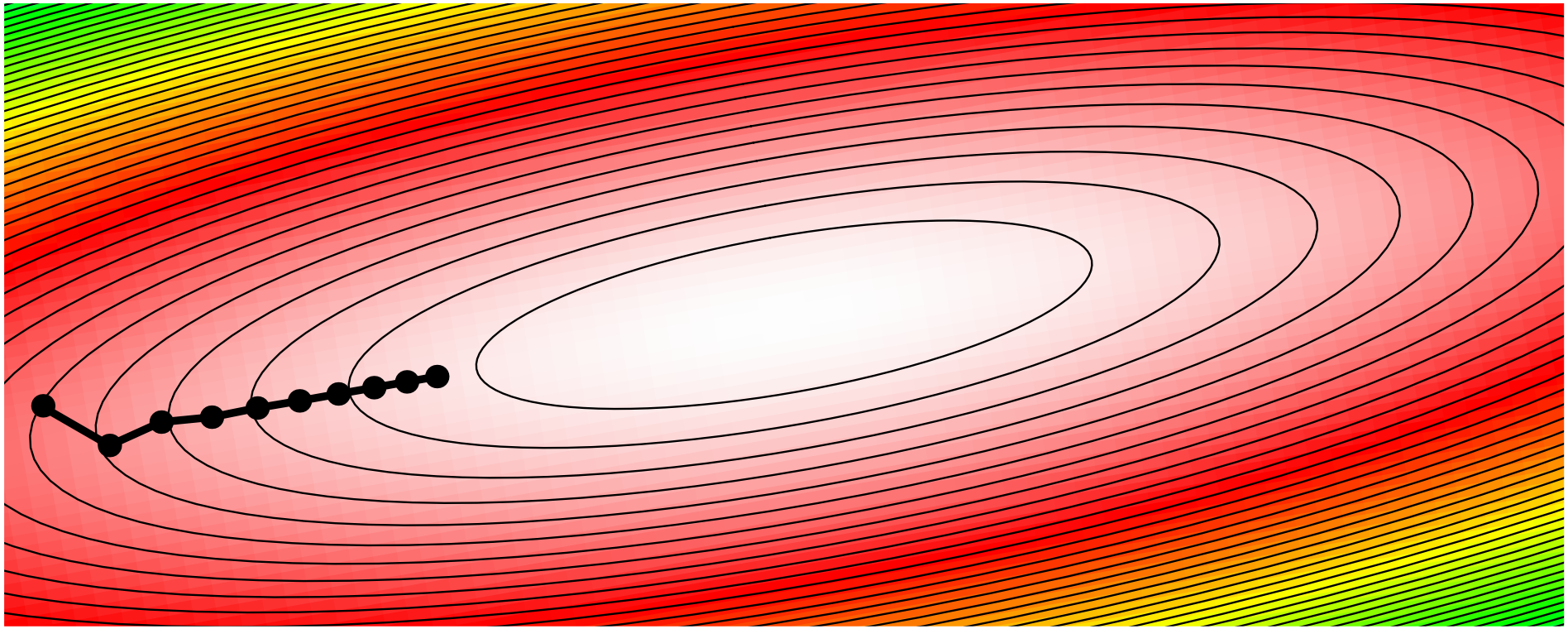

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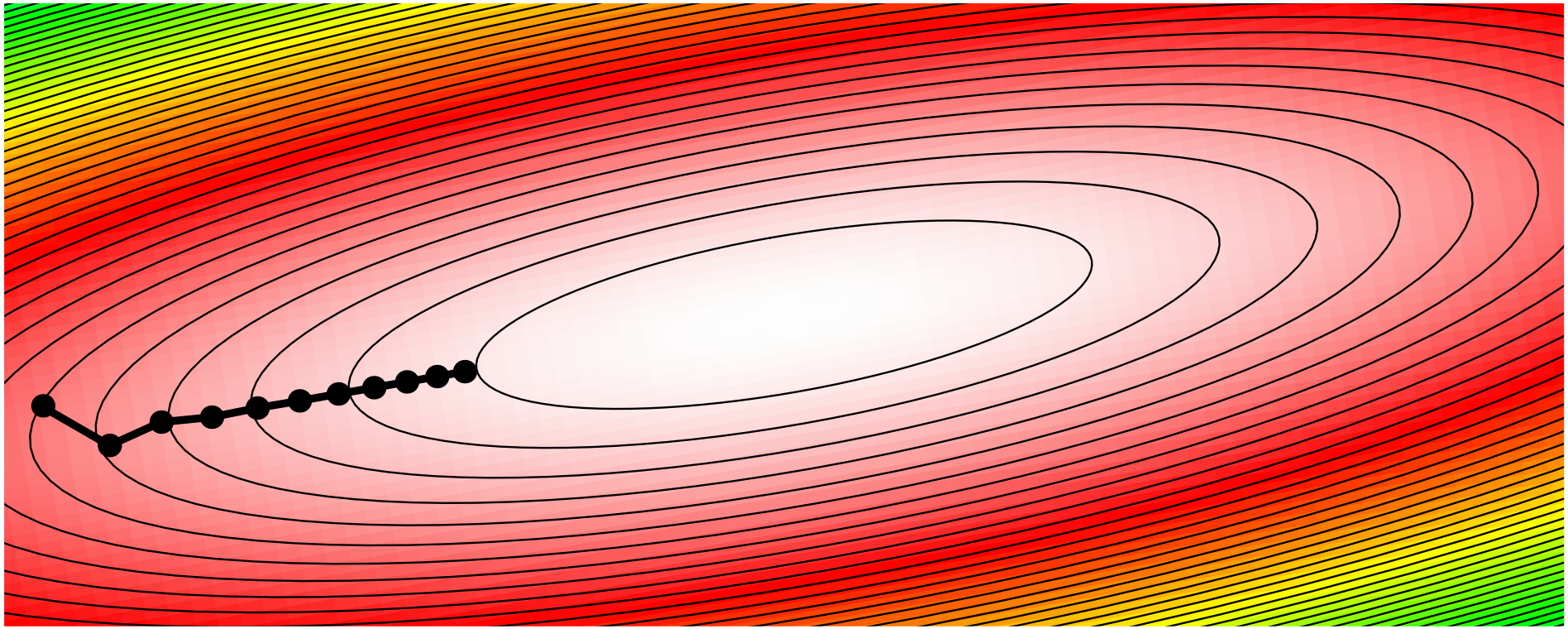

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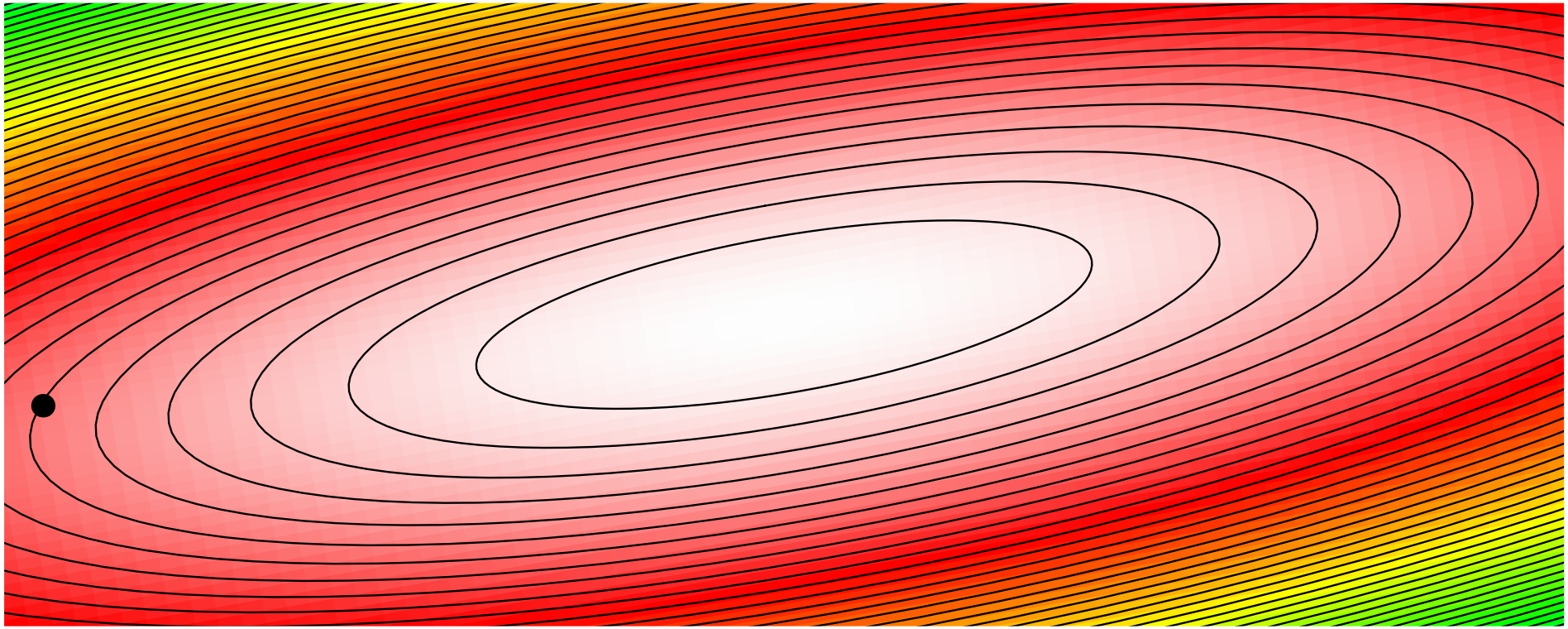

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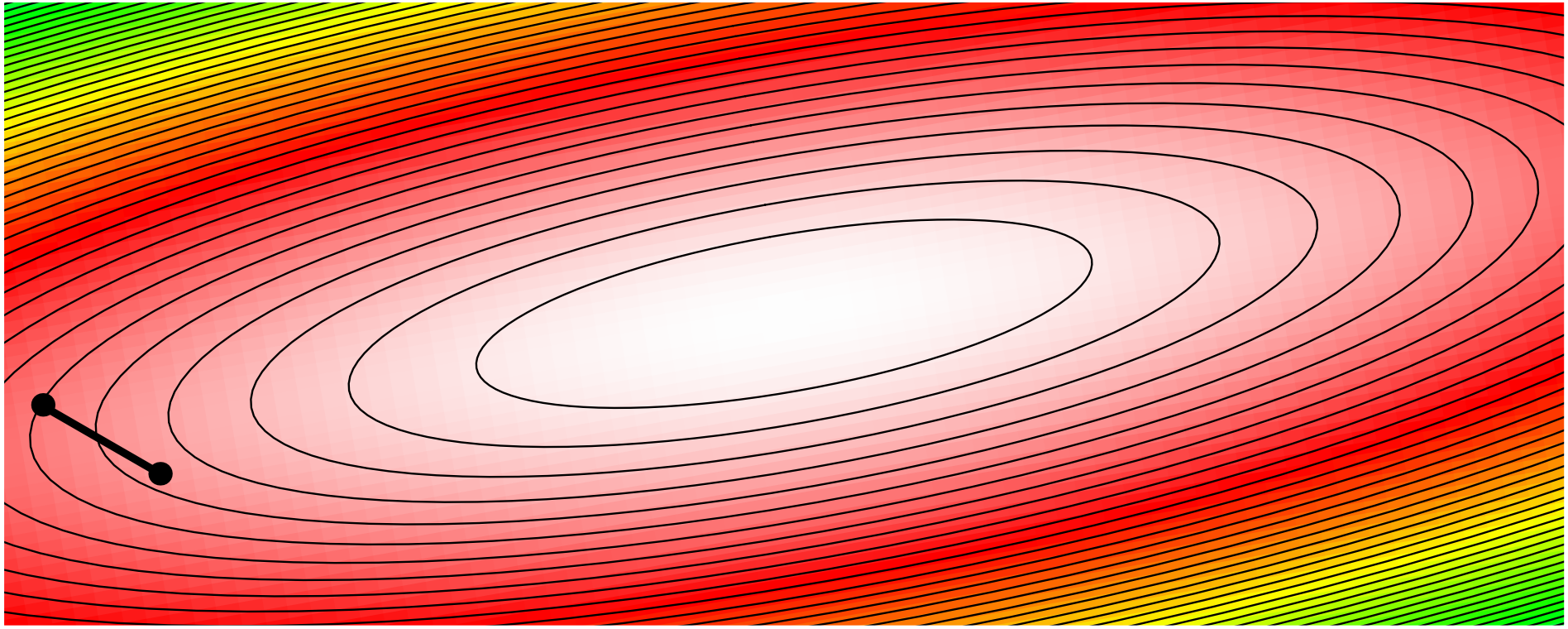

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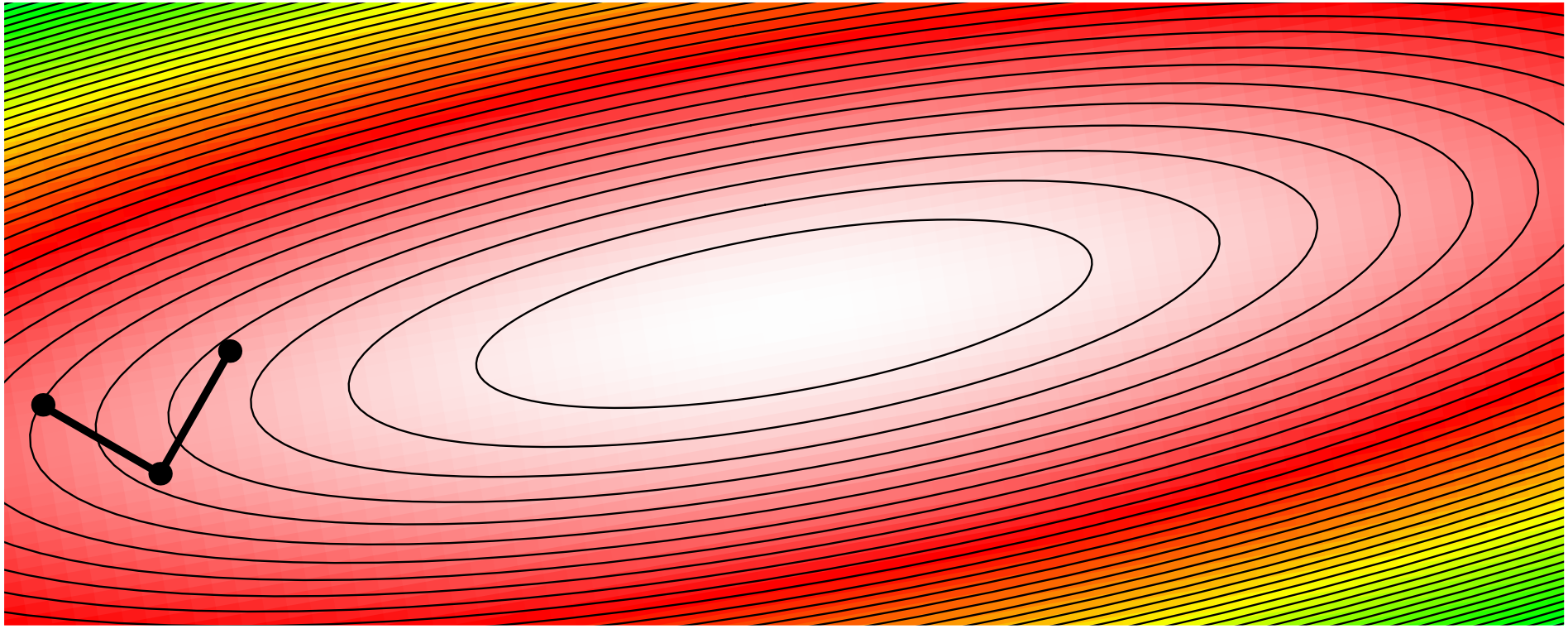

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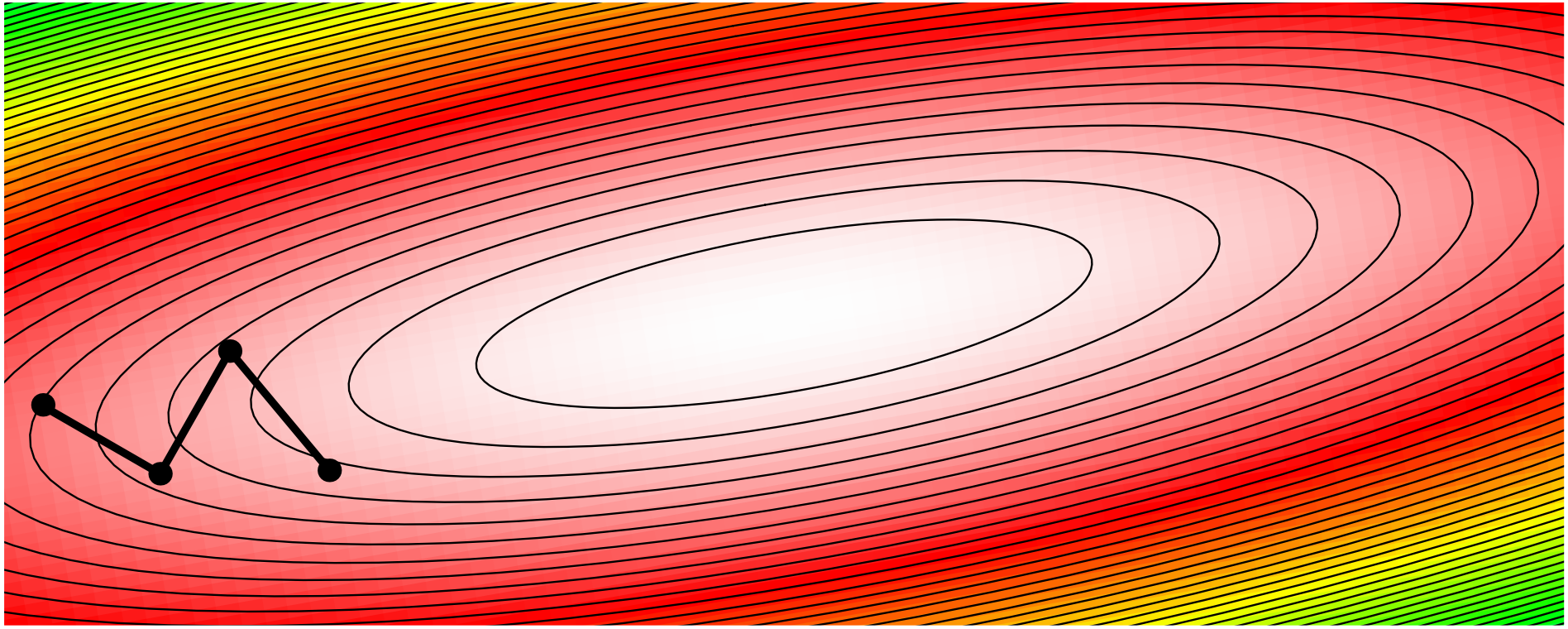

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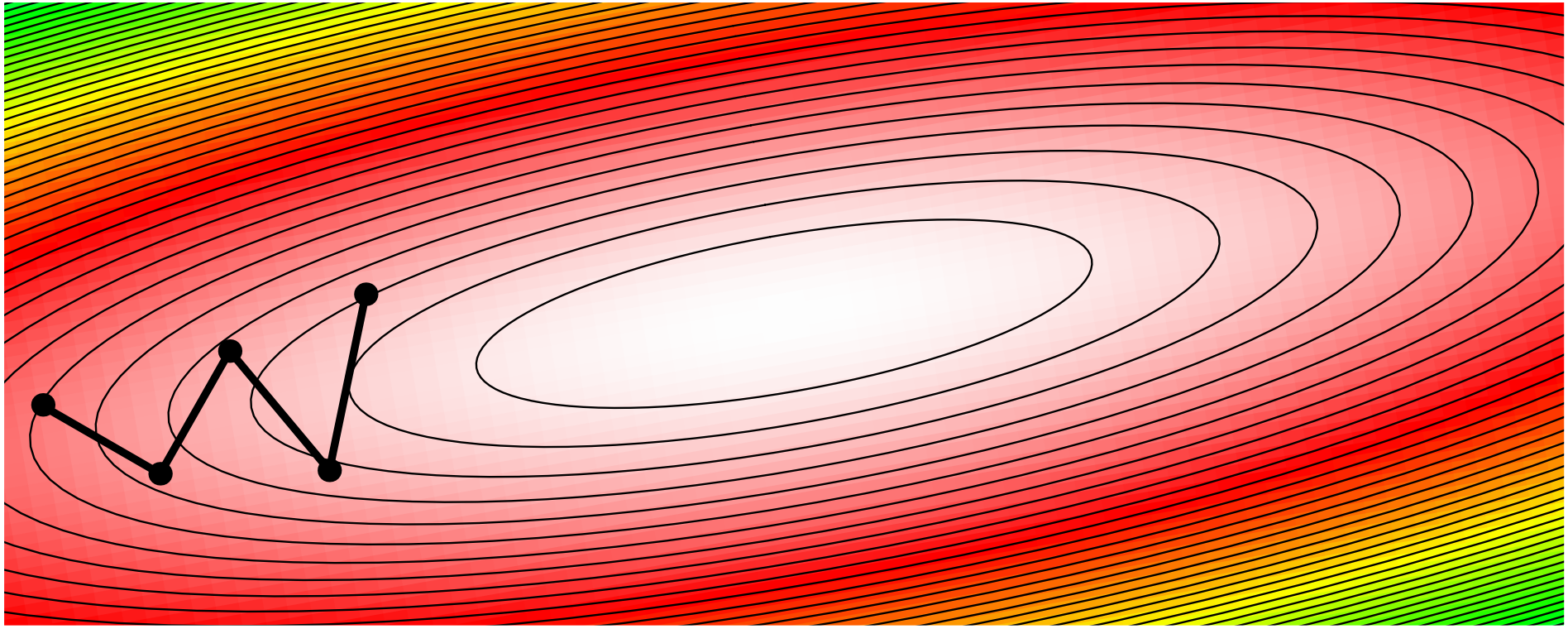

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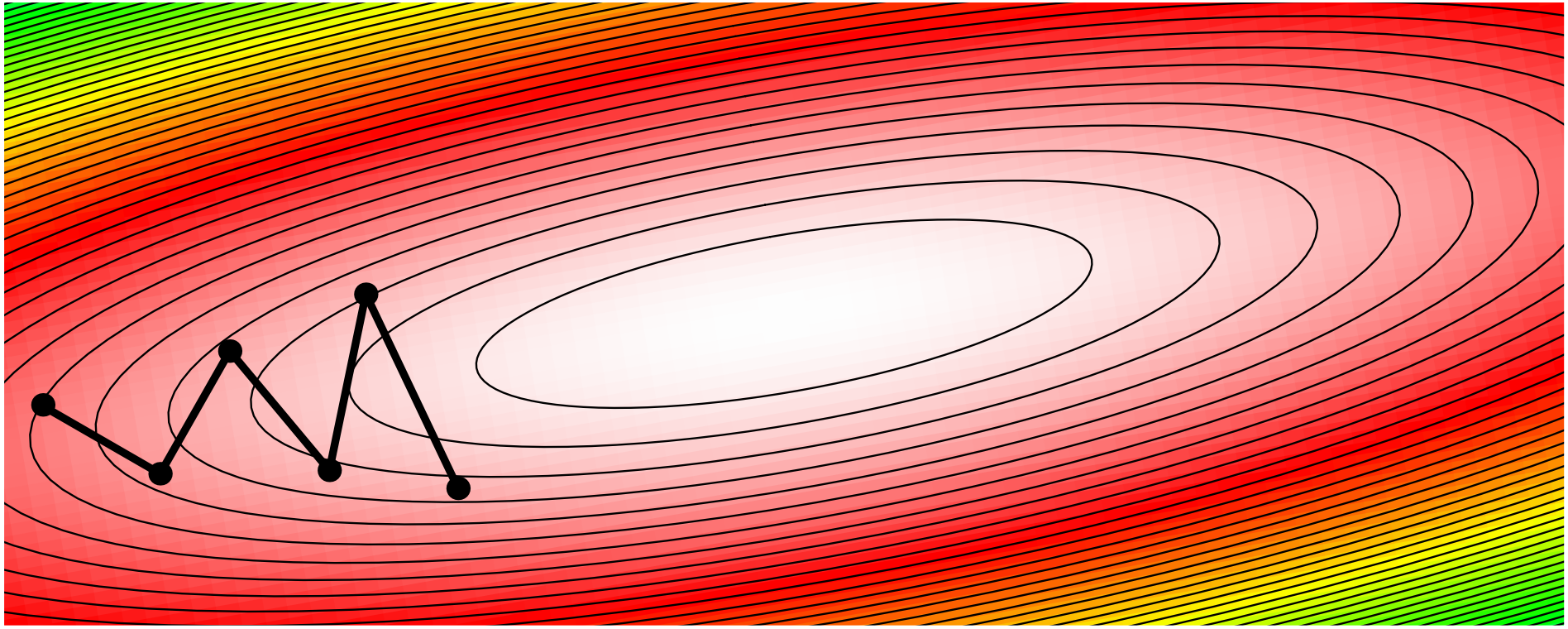

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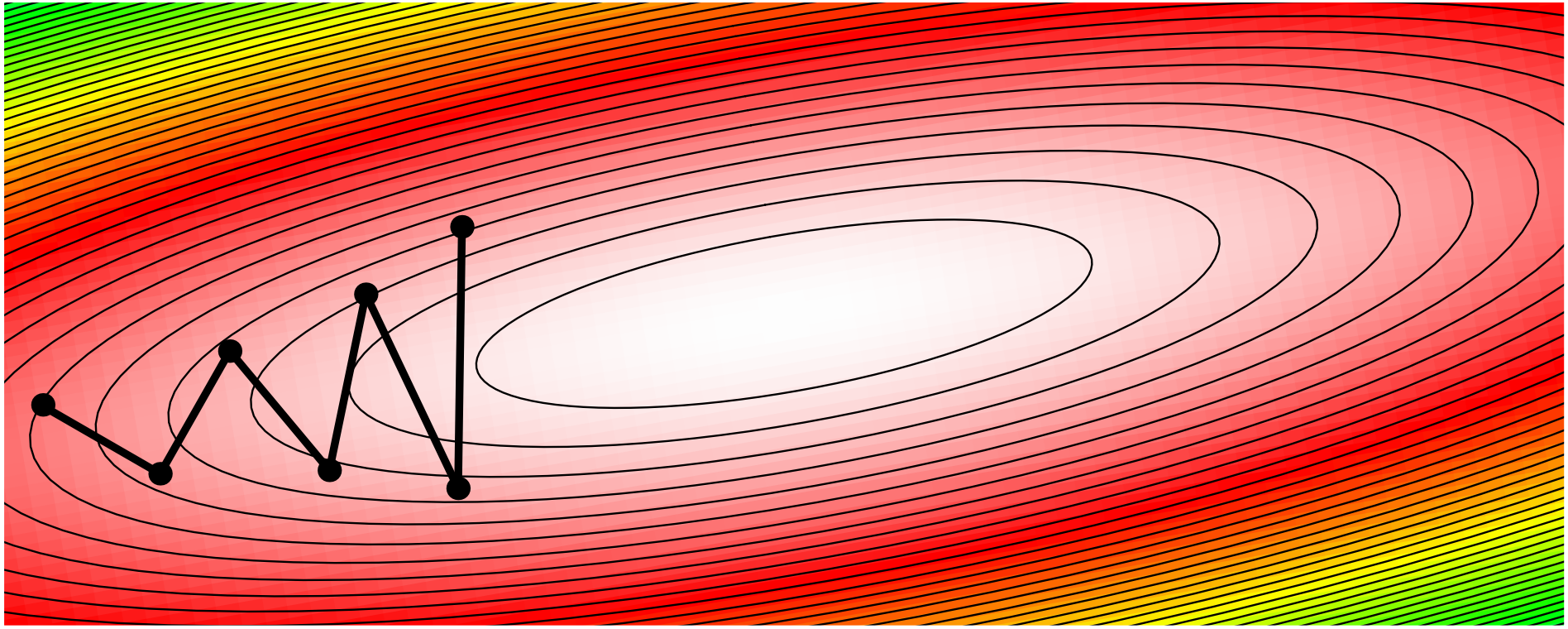

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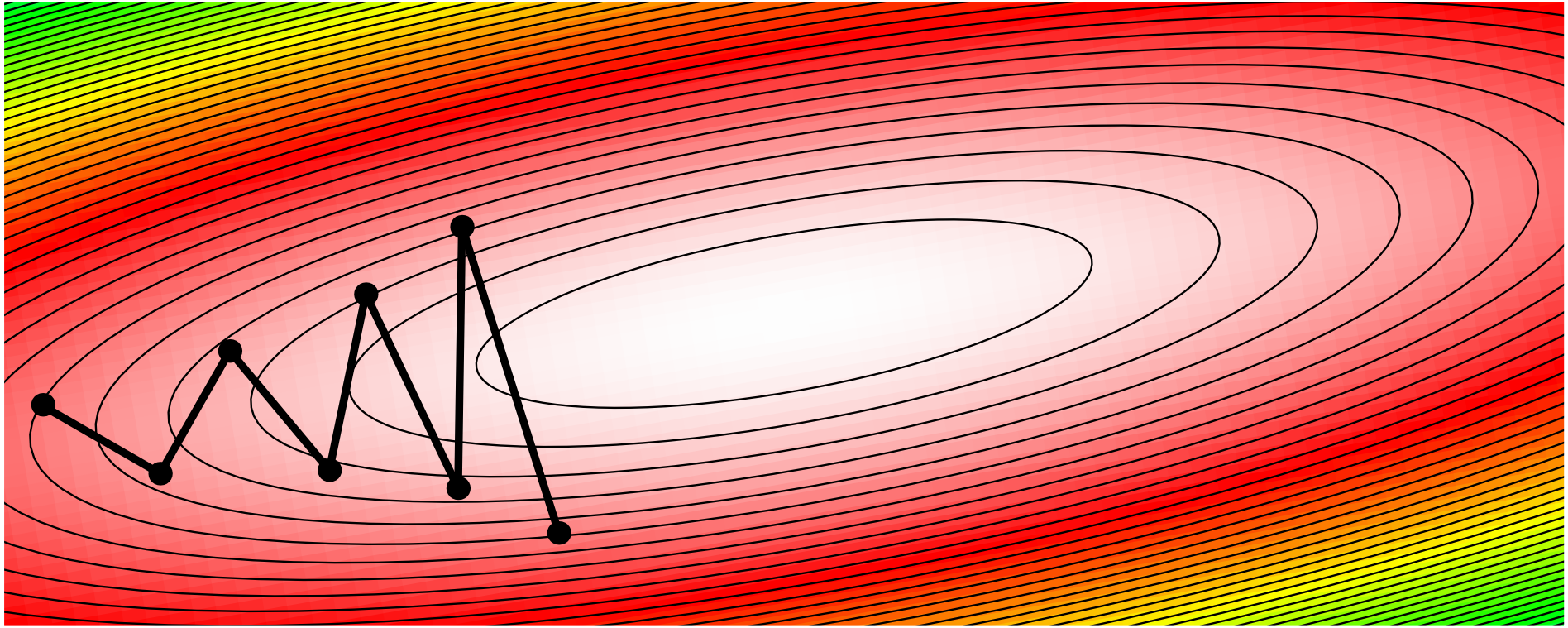

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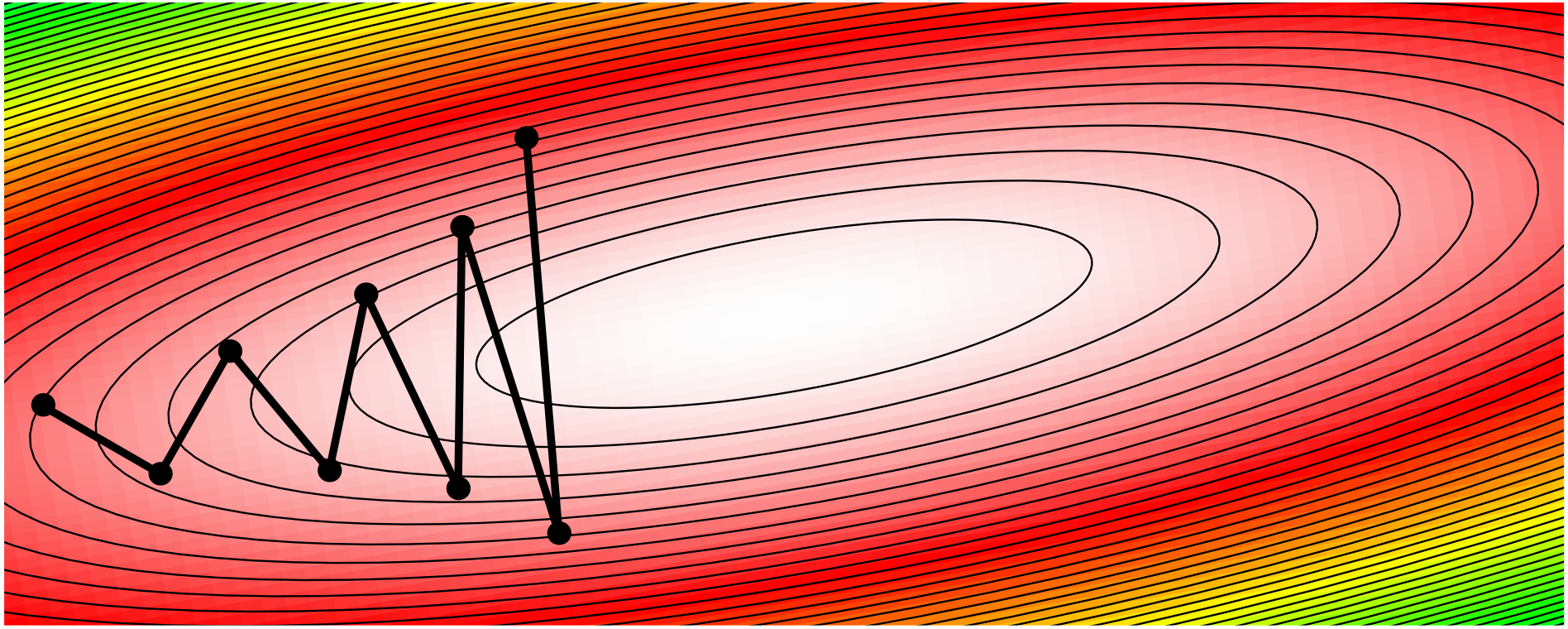

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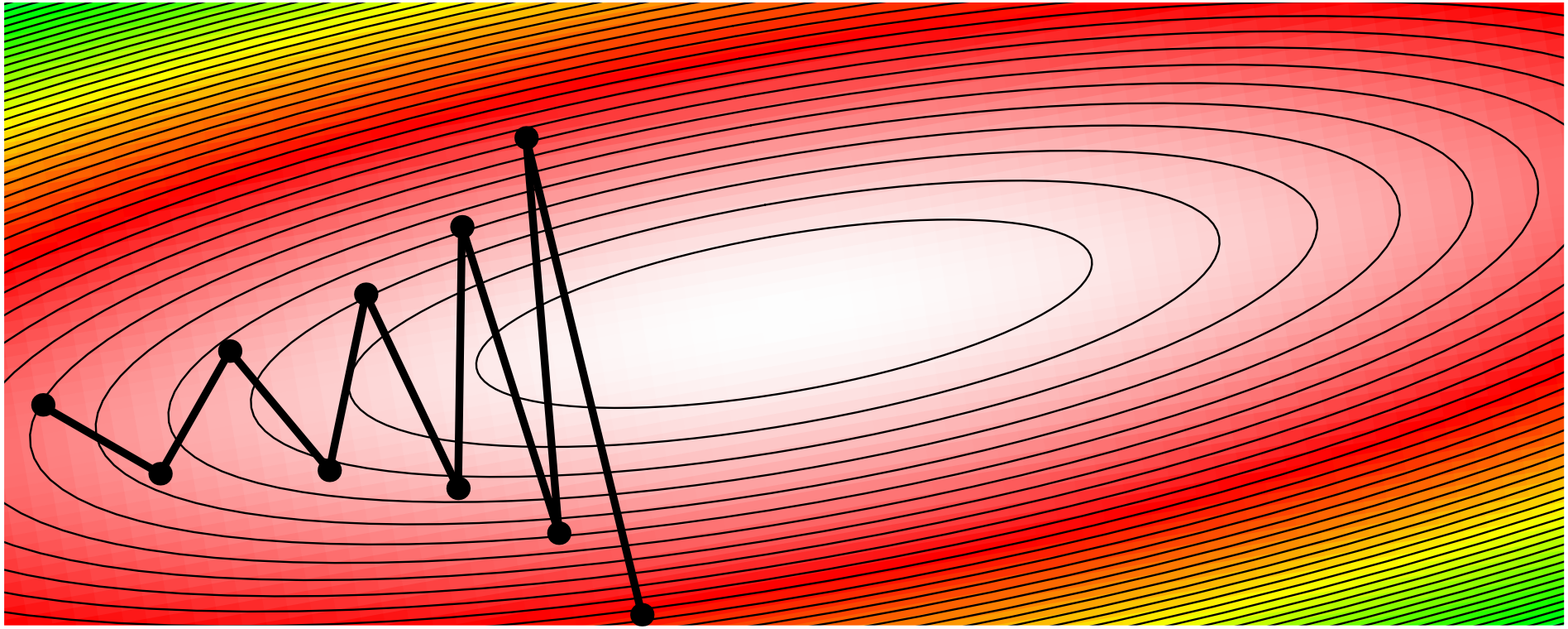

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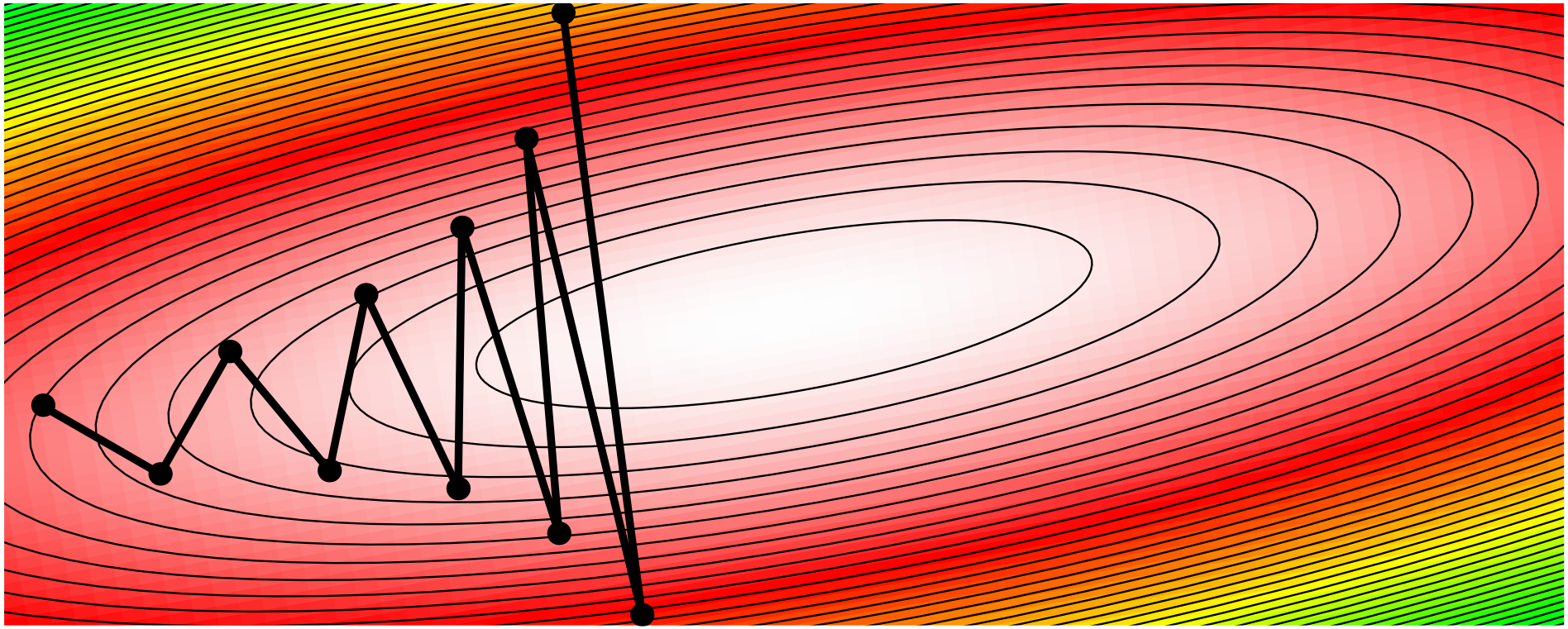

Speeding Up Stochastic Gradient Descent

- To speed up stochastic gradient descent we can add a momentum term

$$\boldsymbol{v}(t) = (1-\epsilon)\boldsymbol{v}(t) - r\boldsymbol{\nabla} L(\boldsymbol{\theta}(t-1))$$
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- Many variations on this, e.g. ada-delta and adam

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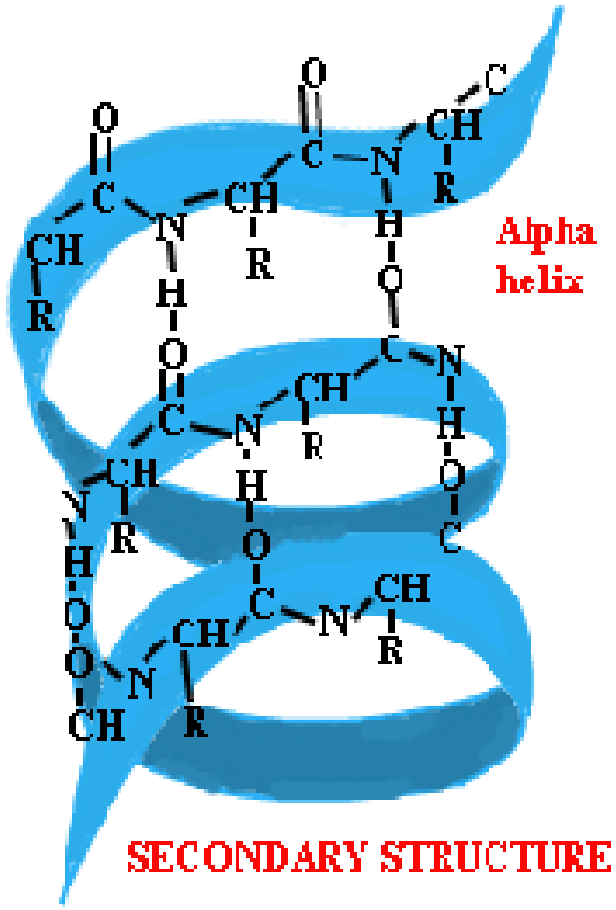
Outline

1. MLP
2. **Secondary Protein Structure**
3. Designing MLPs



Protein Structure

- Understanding the 3D structure of proteins is very important to understand its function
- Unfortunately, it is extremely difficult to infer its 3D structure
  - ⋆ It is both experimentally very costly
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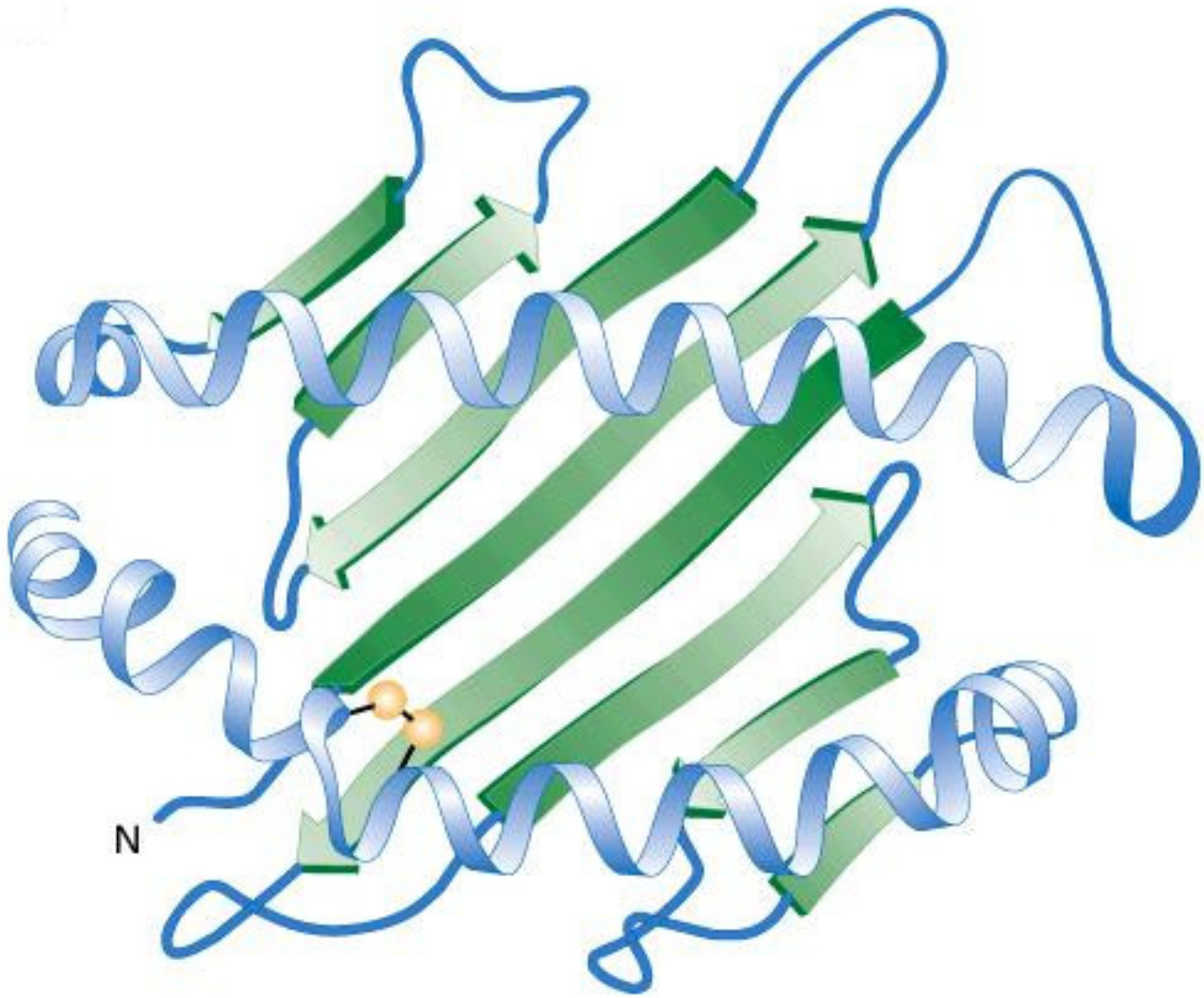

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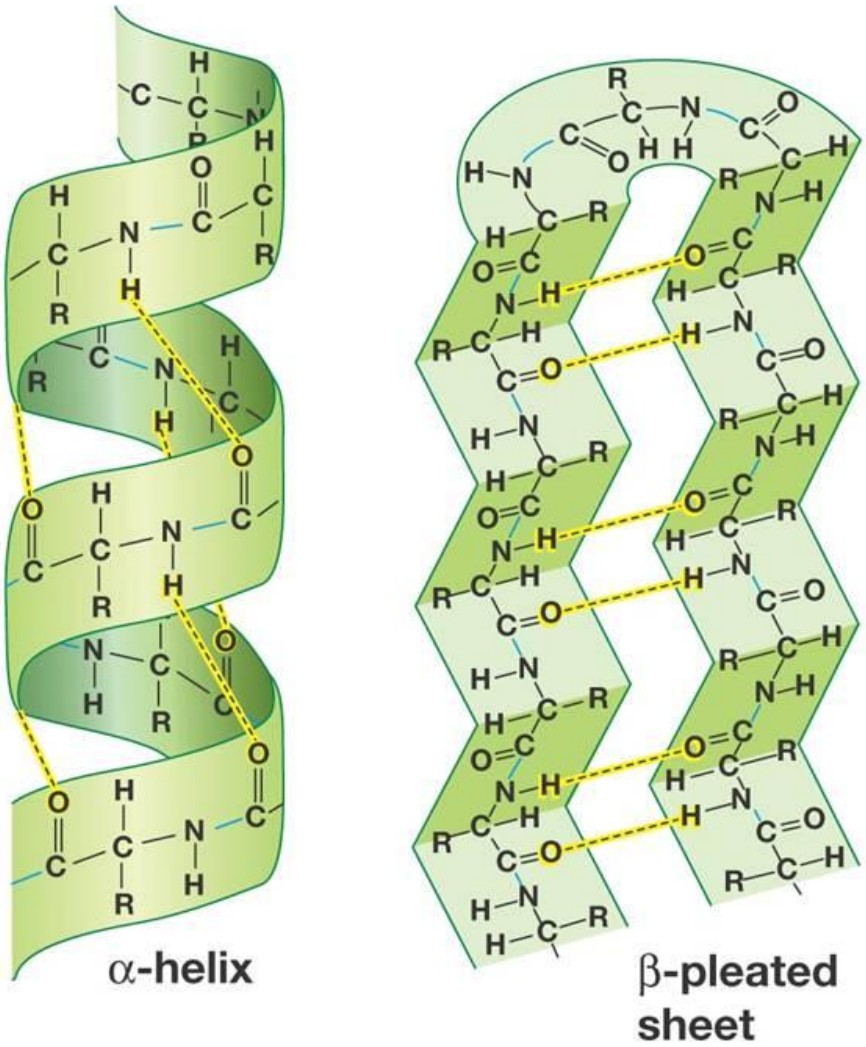

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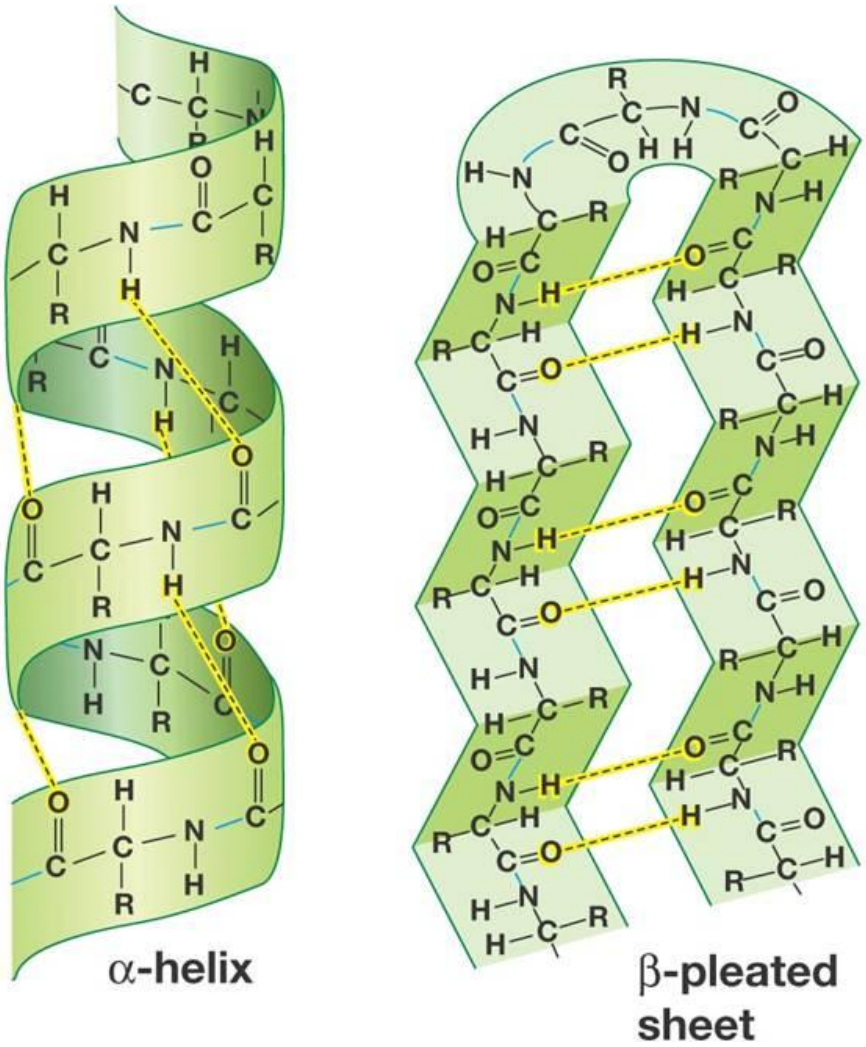

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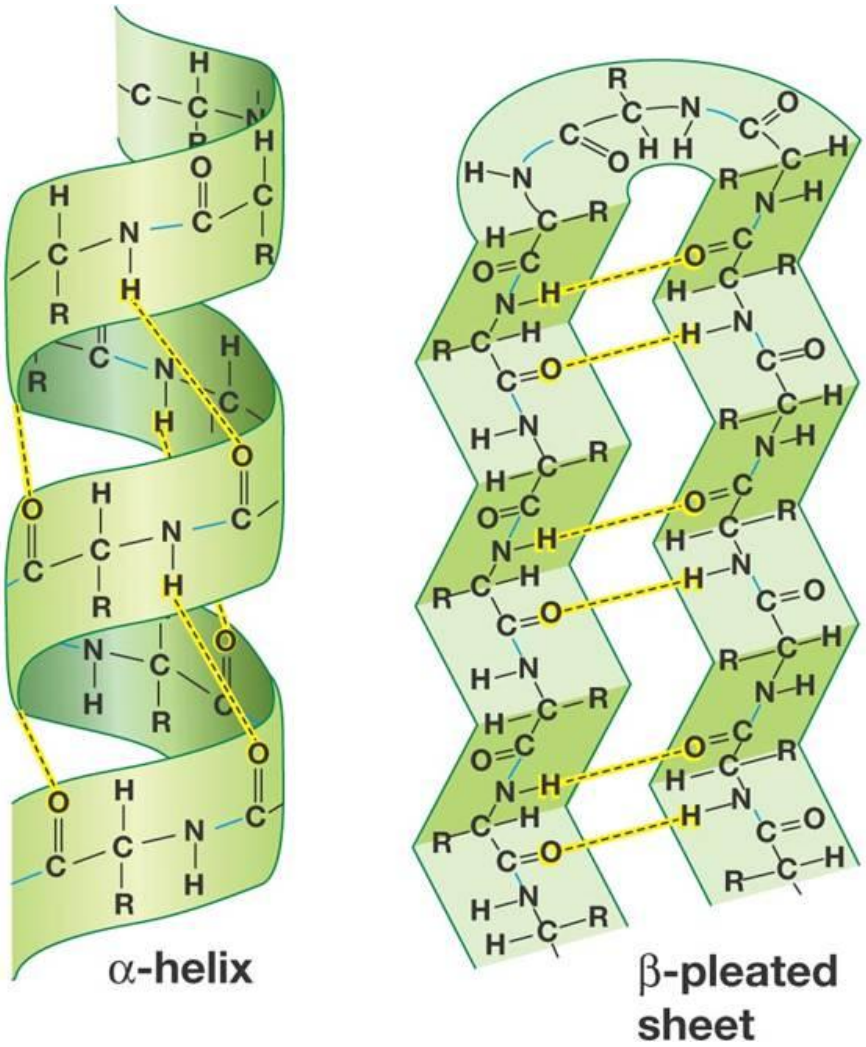

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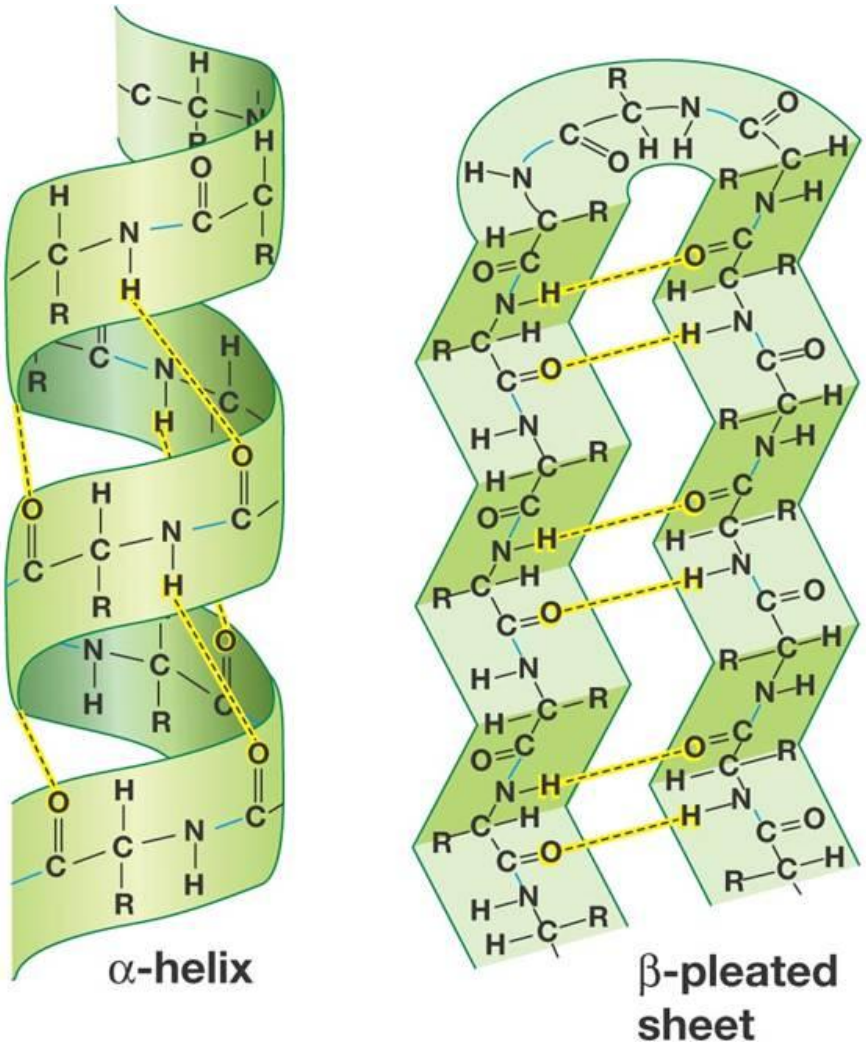

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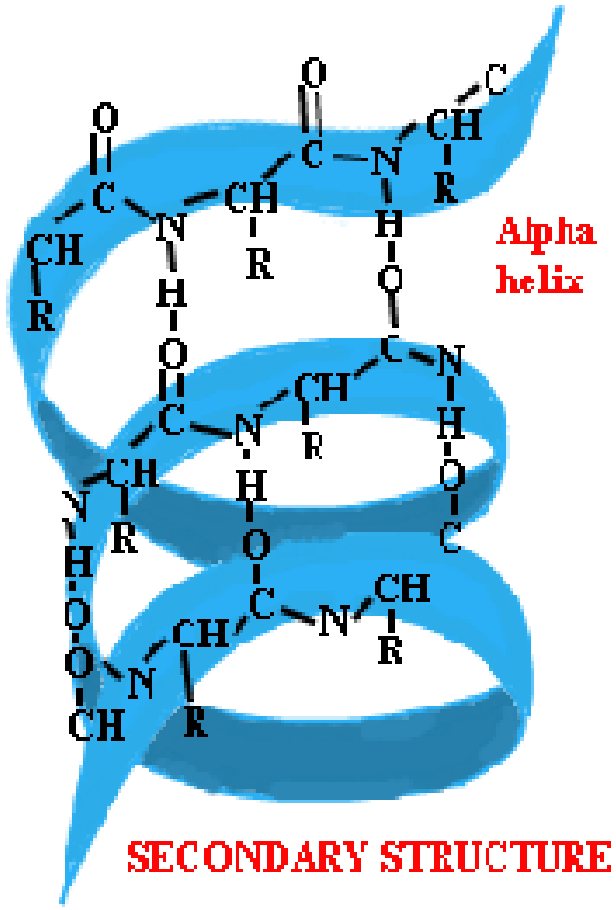

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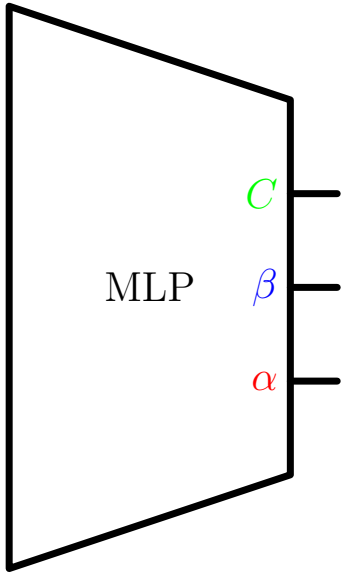

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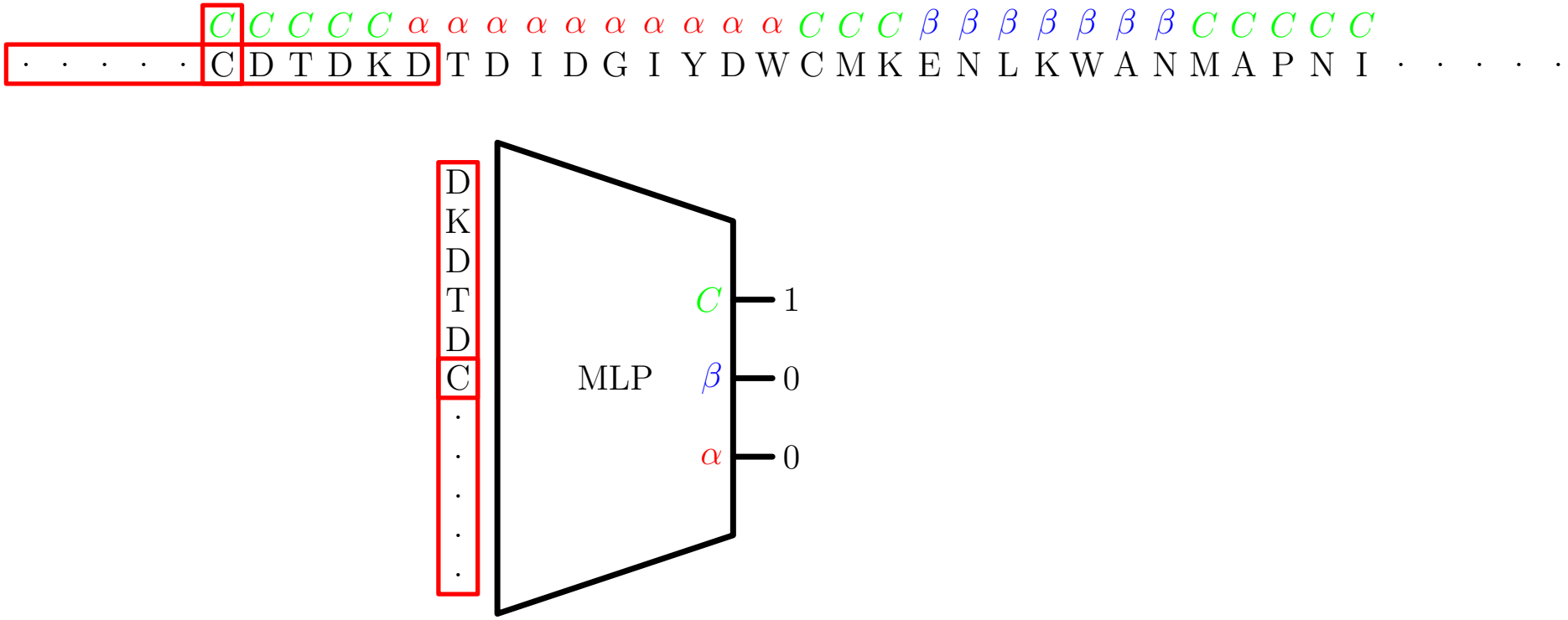
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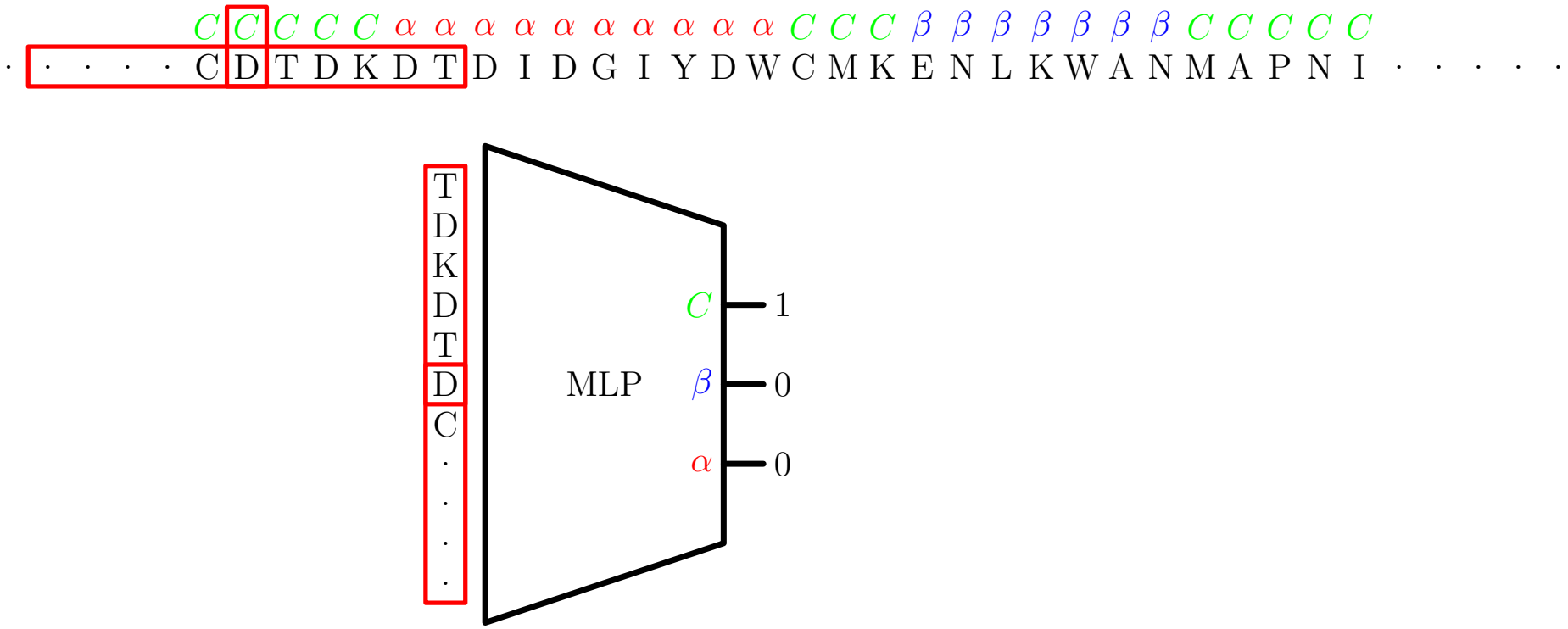
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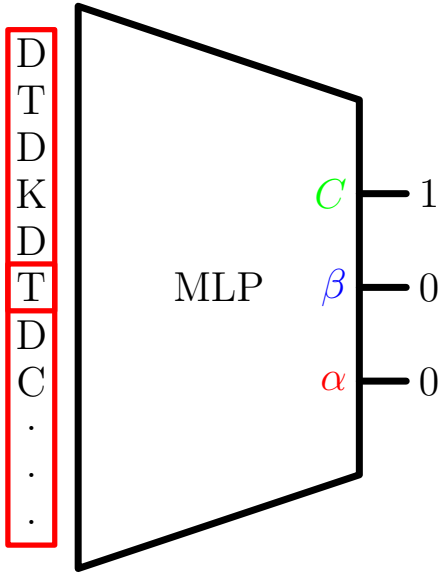

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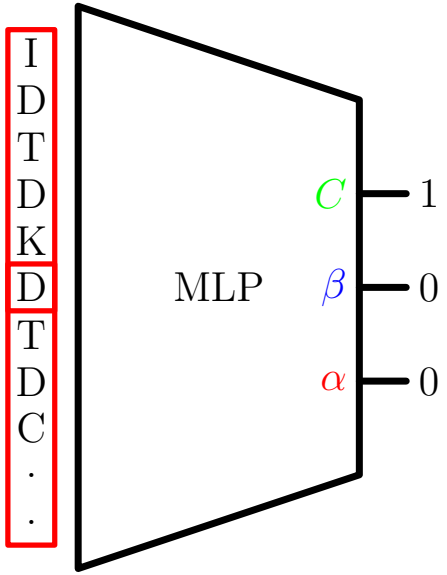

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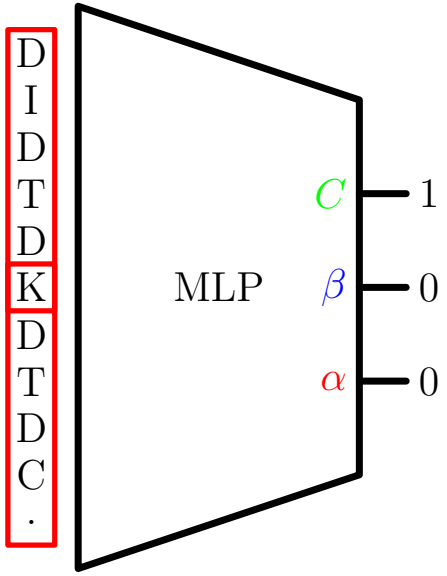

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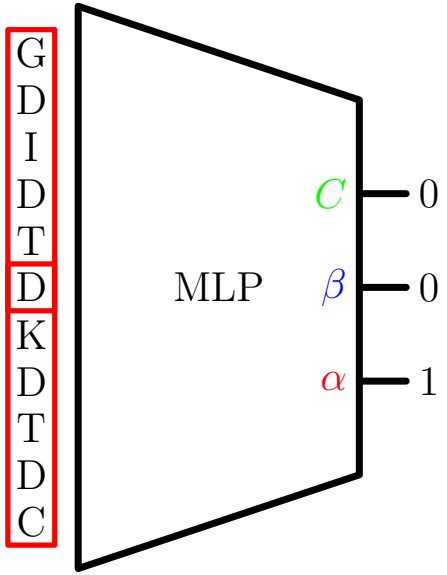

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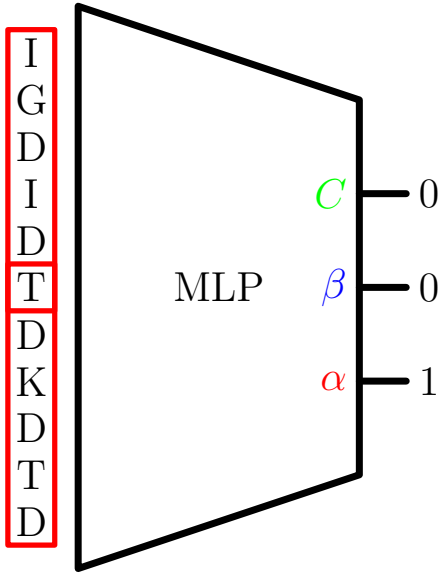

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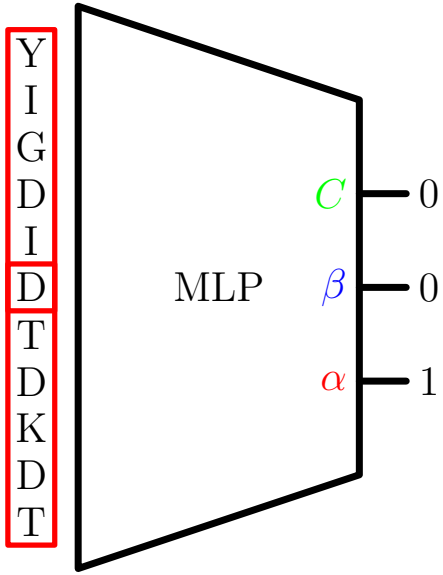

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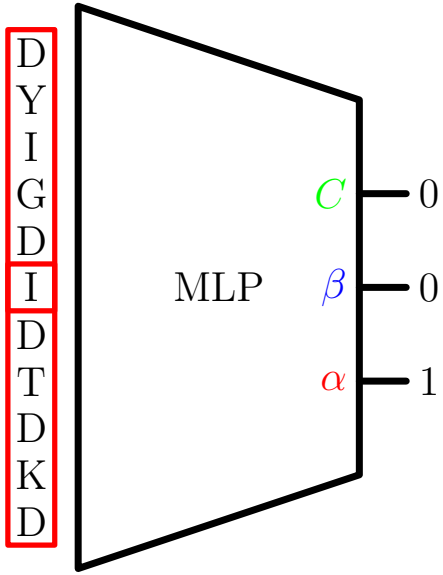

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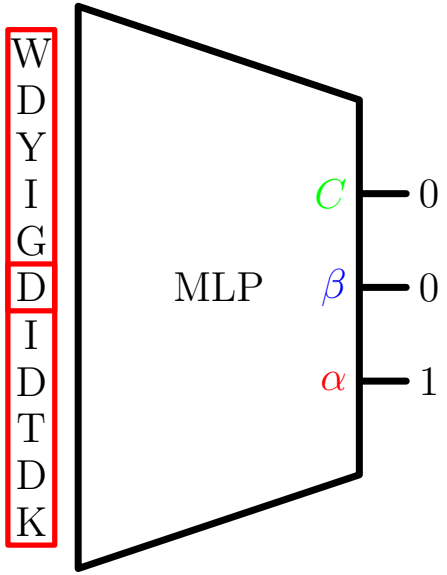

Randomised Windows

- During training we can randomise the placement of the window so we don't learn a long section of one structure

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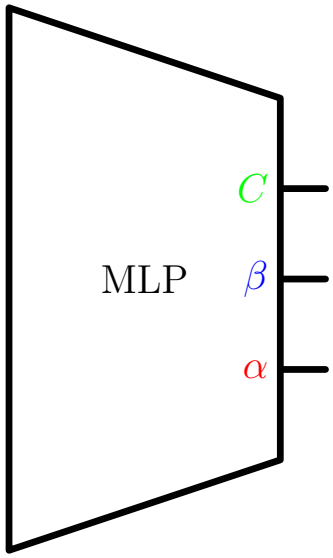

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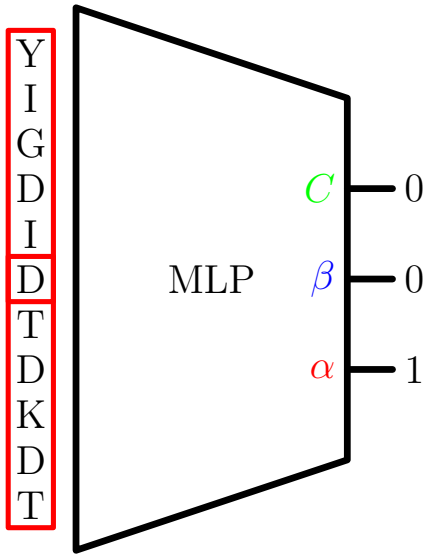

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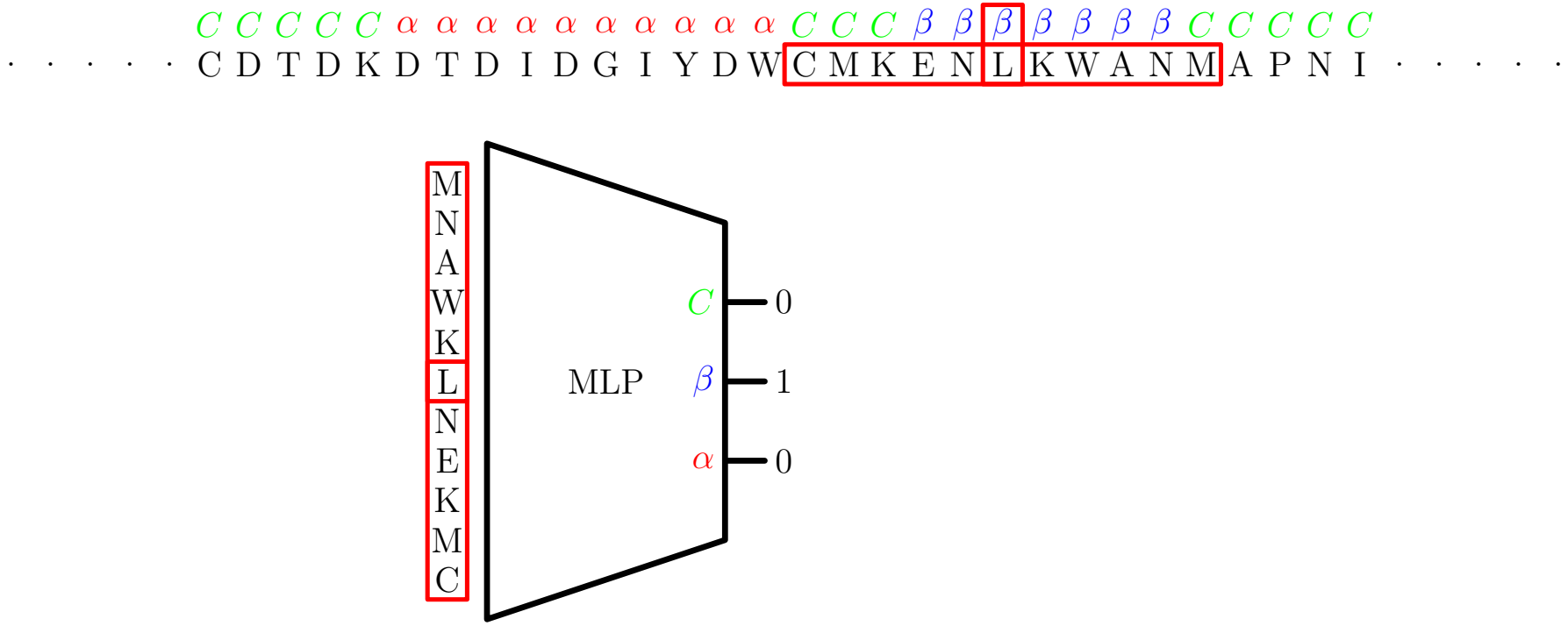

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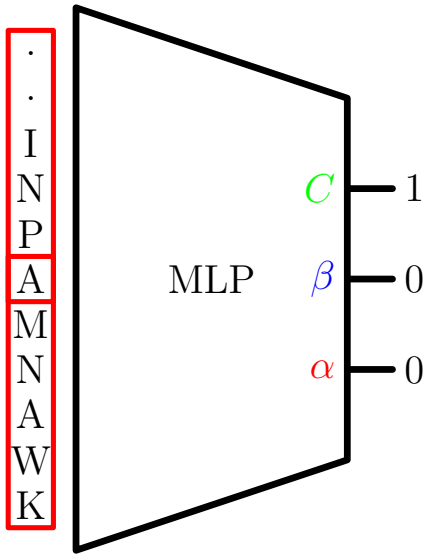

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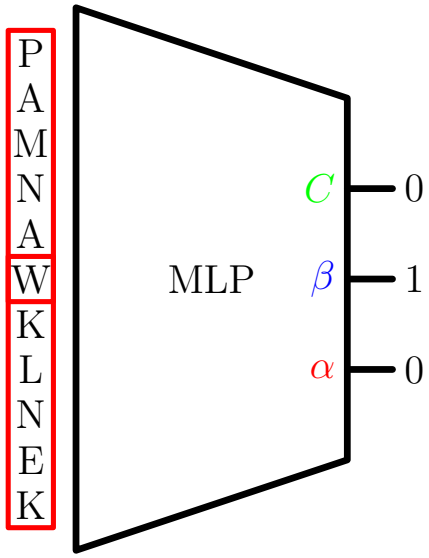

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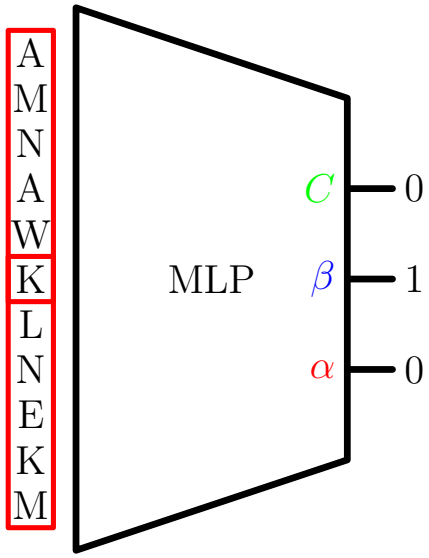

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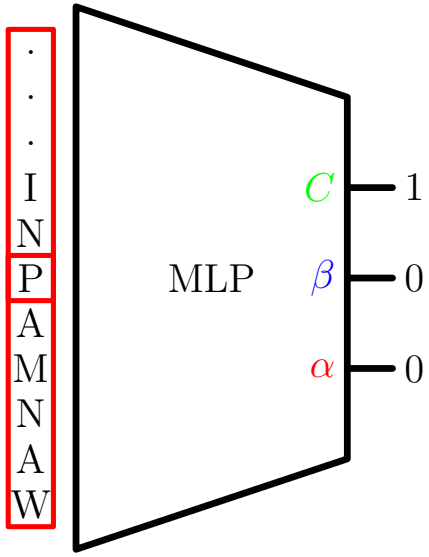

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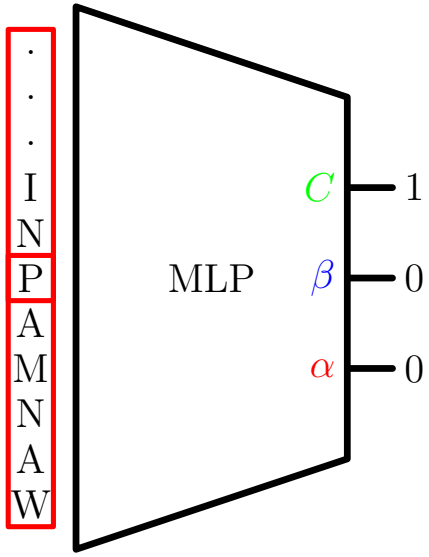

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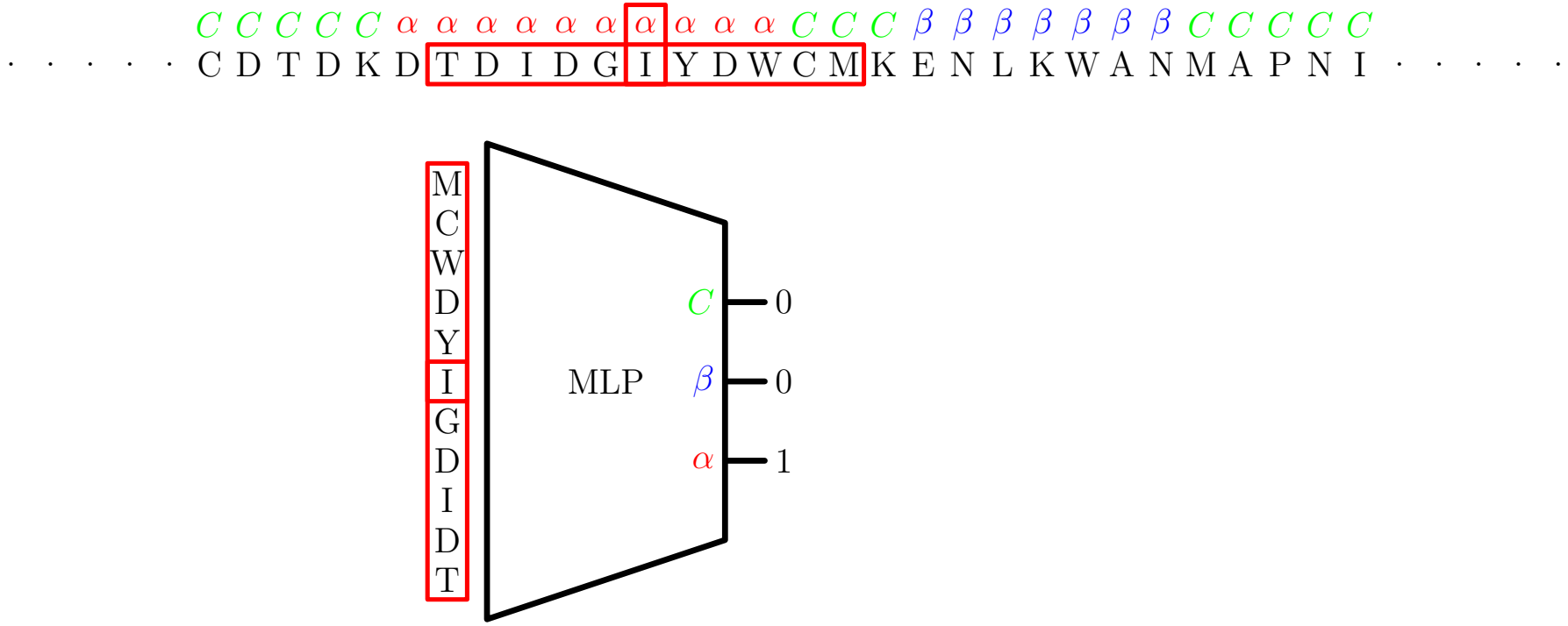

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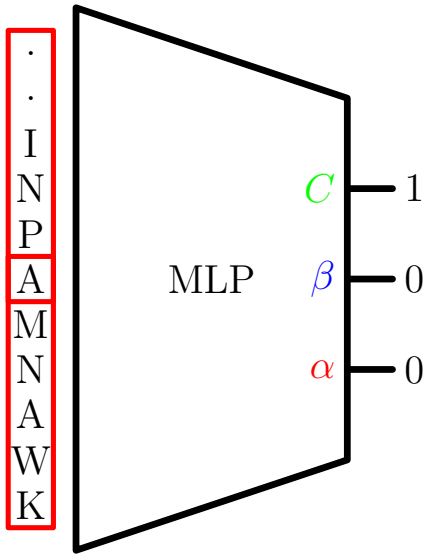

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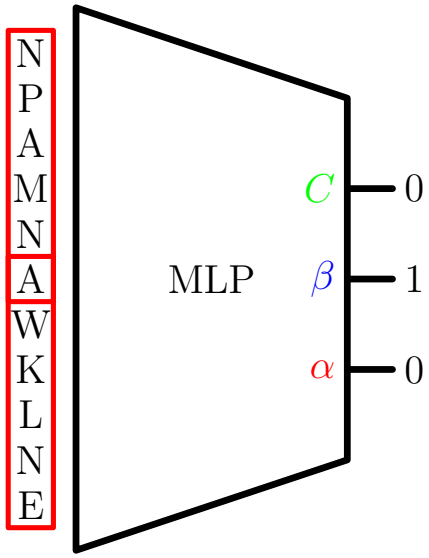

Representing Proteins

- How should we feed our protein to an MLP?
- Providing a number from 1–20 would put in an artificial ordering
- We could try and put in chemical properties, e.g. acid/base hydophobic/hydophilic, etc. but which ones
- It turned out best to use a **1-hot** (orthogonal) representation

$$\begin{aligned}
\text{Glycine (G)} &= (1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0) \\
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\text{End} &= (0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1)
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- This is unbiased and seems to work best

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- Again use a 1-hot representation
- Use winner-takes-all prediction (largest output)

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$$f_i(\boldsymbol{x}|\boldsymbol{\theta}) = \frac{\mathrm{e}^{\sum_{j=1}^{h} \bar{W}_{ij} h_j(\boldsymbol{x}) + \bar{b}_i}}{\sum_{k=1}^{3} \mathrm{e}^{\sum_{j=1}^{h} \bar{W}_{kj} h_j(\boldsymbol{x}) + \bar{b}_i}}$$

- where $h_j(\boldsymbol{x})$ is the output of the jth hidden unit

- Note that we can the interpret the output as

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$$= \sum_{\mu=1}^{P} \log(f_{y^\mu}(\boldsymbol{x}^\mu|\boldsymbol{\theta}))$$

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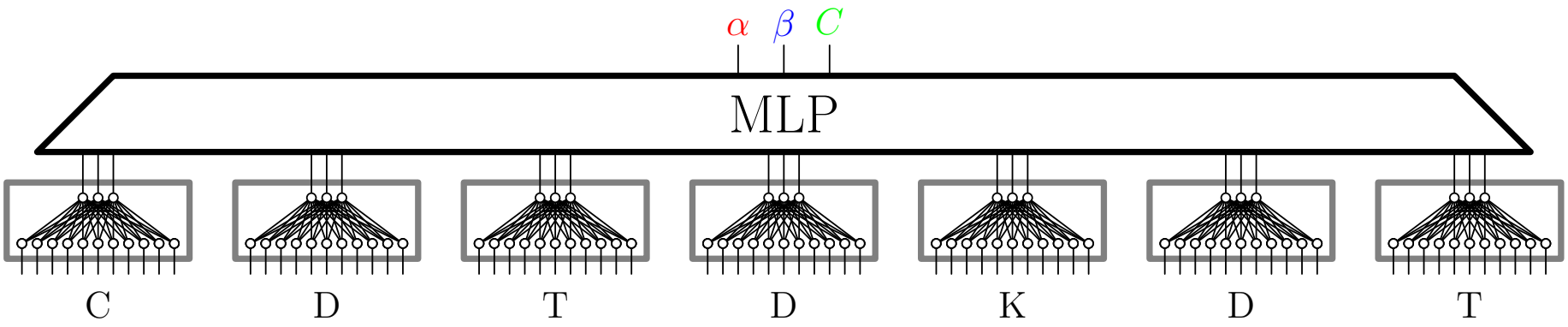
- The 1-hot vector representation of the input makes the network very large

- We can learn an encoding or **embedding** by using the same 21 to M layer for all inputs in the window



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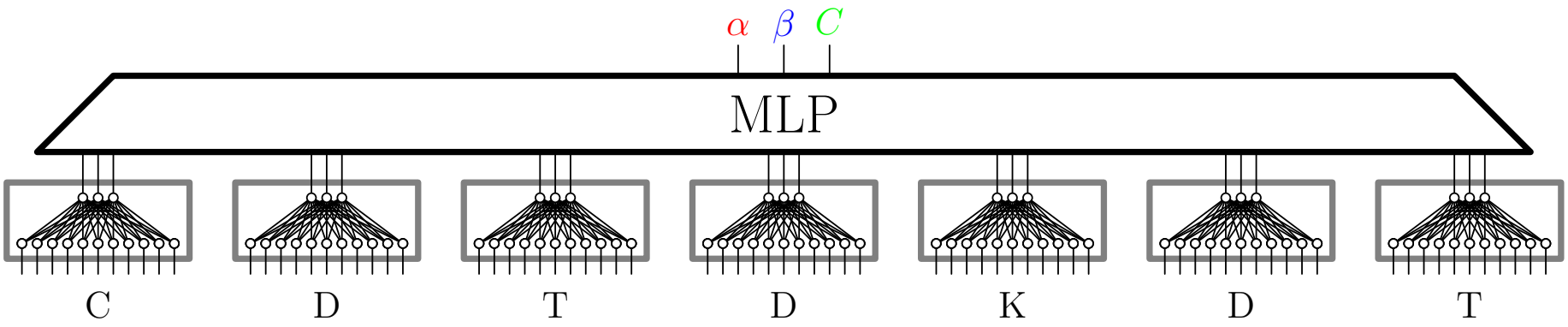

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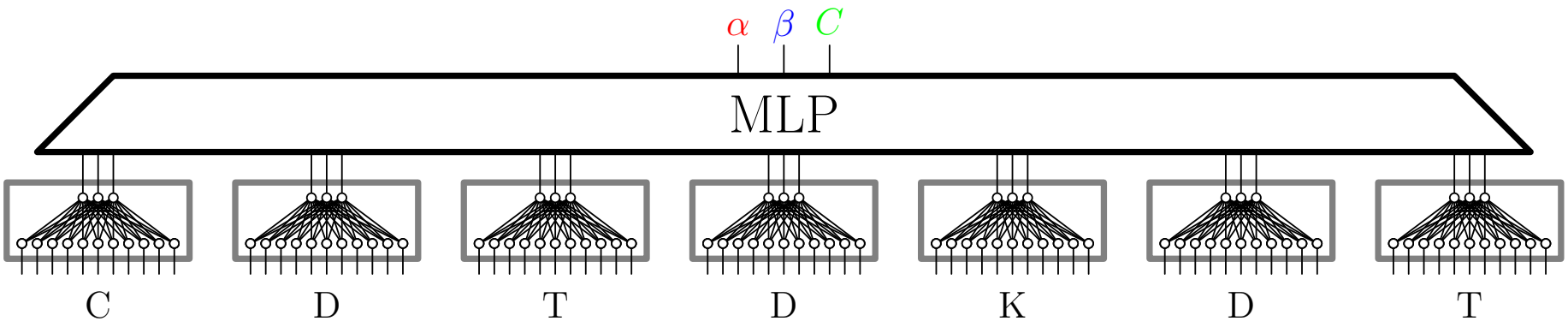

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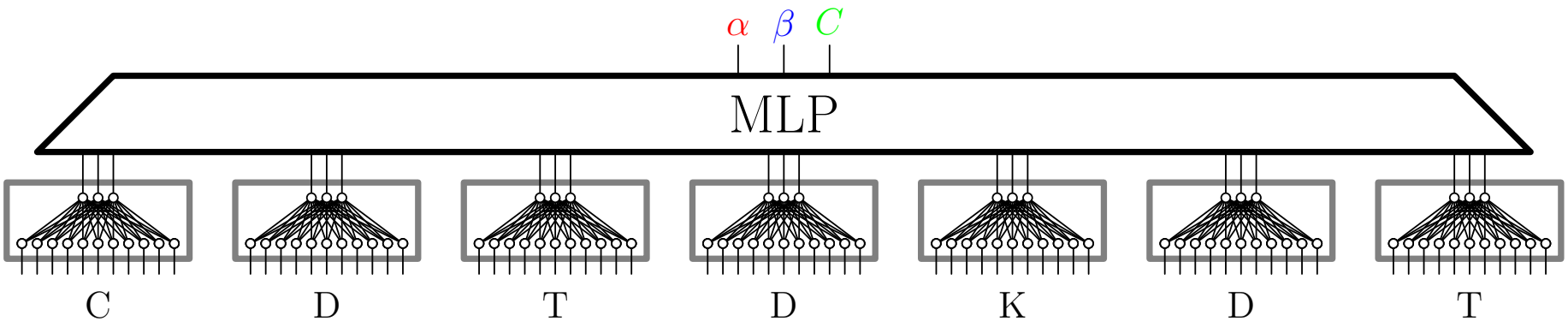

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- We can improve our machine by building a second machine that takes the prediction at a site and its neighbours
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Using Profiles

- The big improvement in prediction came from using the protein profile obtained by finding similar protein structures

```
A5ASC3.1    14 SIKLWPPSQTTRLLLVERMANNLST..PSIFTRK..YGSLSKEEARENAKQIEEVACSTANQ.....HYEKEPDGDGGSAVQLYAKECSKLILEVLK 101
B4F917.1    13 SIKLWPPSESTRIMLVDRMTNNLST..ESIFSRK..YRLLGKQEAHENAKTIEELCFALADE.....HFREEPDGDGSSAVQLYAKETSKMMLEVLK 100
A9S1V2.1    23 VFKLWPPSQGTREAVRQKMALKLSS..ACFESQS..FARIELADAQEHARAIEEVAFGAAQE......ADSGGDKTGSAVVMVYAKHASKLMLETLR 109
B9GSN7.1    13 SVKLWPPGQSTRLMLVERMTKNFIT..PSFISRK..YGLLSKEEAEEDAKKIEEVAFAAANQ.....HYEKQPDGDGSSAVQIYAKESSRLMLEVLK 100
Q8H056.1    30 SFSIWPPTQRTRDAVVRRLVDTLGG..DTILCKR..YGAVPAADAEPAARGIEAEAFDAAAA..SGEAAATASVEEGIKALQLYSKEVSRRLLDFVK 120
Q0D4Z3.2    44 SLSIWPPSQRTRDAVVRRLVQTLVA..PSILSQR..YGAVPEAEAGRAAAAVEAEAYAAVTES.SSAAAAPASVEDGIEVLQAYSKEVSRRLLELAK 135
B9MVW8.1    56 SFSIWPPTQRTRDAIISRLIETLST..TSVLSKR..YGTIPKEEASEASRRIEEEAFSGAST.......VASSEKDGLEVLQLYSKEISKRMLETVK 141
Q0IYC5.1    29 SFAVWPPTRRTRDAVVRRLVAVLSGDTTTALRKRYRYGAVPAADAERAARAVEAQAFDAASA....SSSSSSSVEDGIETLQLYSREVSNRLLAFVR 121
A9NW46.1    13 SIKLWPPSESTRLMLVERMTDNLSS..VSFFSRK..YGLLSKEEAAENAKRIEETAFLAAND.....HEAKEPNLDDSSVVQFYAREASKLMLEALK 100
Q9C500.1    57 SLRIWPPTQKTRDAVLNRLIETLST..ESILSKR..YGTLKSDDATTVAKLIEEEAYGVASN.......AVSSDDDGIKILELYSKEISKRMLESVK 142
Q2HRI7.1    25 MYSIWPPKQRTRDAVKNRLIETLST..PSVLTKR..YGTMSADEASAAAIQIEDEAFSVANA.......SSSTSNDNVTILEVYSKEISKRMIETVK 110
Q9M7N3.1    28 SFKIWPPTQRTREAVVRRLVETLTS..QSVLSKR..YGVIPEEDATSAARIIEEEAFSVASV.ASAASTGGRPEDEWIEVLHIYSQEIXQRVVESAK 119
Q9M7N6.1    25 SFSIWPPTQRTRDAVINRLIESLST..PSILSKR..YGTLPQDEASETARLIEEEAFAAAGS.......TASDADDGIEILQVYSKEISKRMIDTVK 110
Q9LE82.1    14 SVKMWPPSKSTRLMLVERMTKNITT..PSIFSRK..YGLLSVEEAEQDAKRIEDLAFATANK.....HFQNEPDGDGTSAVHVYAKESSKLMLDVIK 101
Q9M651.2    13 SIKLWPPSLPTRKALIERITNNFSS..KTIFTEK..YGSLTKDQATENAKRIEDIAFSTANQ.....QFEREPDGDGGSAVQLYAKECSKLILEVLK 100
B9R748.1    48 SLSIWPPTQRTRDAVITRLIETLSS..PSVLSKR..YGTISHDEAESAARRIEDEAFGVANT.......ATSAEDDGLEILQLYSKEISRRMLDTVK 133
```

Profile Encoding

- Instead of using the protein sequence as input we use a coding showing all possible substitutions, deletions or insertions at a site
- This works because protein structure is much better conserved than protein sequences
- Thus related protein sequences will with high probability have the same secondary structure
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